Constrained-Optimal Social Security Reform with a Debt Ceiling and Learning by Doing

HongSeok Kim*

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Abstract

Once the Social Security trust fund is depleted, all bets are off. The most likely scenario is what I call the baseline or no-reform case: an initial transition of increasing government transfers to finance the Social Security deficit, which would increase national debt up to a ceiling, followed by a combination of higher payroll taxes and lower benefits. Can a different adjustment path mitigate this burden without making any generation worse off on average than under the no-reform? I study this transition in a general-equilibrium overlapping-generations model calibrated to the United States in 2010, with learning by doing and a public-debt ceiling of 150% of GDP. The baseline economy finances scheduled benefits through borrowing until the debt ceiling binds and benefit reductions follow. I construct a reform that lowers the payroll tax to 3.5% by 2200 and reduces benefits more relative to baseline, without average welfare losses for any birth cohort. Cohort-specific payroll-tax relief during working life and benefit supplements for low-income individuals late in retirement protect transition cohorts. A consumption tax finances both under the same debt ceiling, at a present-value cost of 2.2% of 2026 GDP. Learning by doing is essential to this result: without it, the gains from lower payroll taxation are insufficient to finance the cohort-specific payroll tax relief and benefit supplements. Complete phase-out produces larger long-run efficiency gains but imposes welfare losses on many cohorts alive today. Retaining a smaller program, as implied by my reform, yields higher social welfare than phase-out when the government counts only cohorts alive today or weights future cohorts lightly through dynastic altruism.

Keywords: Welfare-improving transition, intergenerational redistribution, dynastic altruism.
JEL: H55, H63, E62, D15, J26.

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1 Introduction

Social Security's combined trust fund reserves are projected to run out in 2034.[1] At that point, payroll revenue would cover only about 83% of scheduled benefits (Social Security Trustees, 2026). Without any reform, the ensuing scenario would likely require higher payroll taxes, lower pension benefits, or more borrowing, and each option moves the cost across generations. Higher taxes and lower benefits fall on current workers and retirees, while borrowing shifts part of the burden to future generations.

France illustrates how difficult it is to reform Social Security and the likely scenario that the US should expect in near future. French governments have met mass opposition to reform for three decades. In 1995, three weeks of strikes led the government to abandon its plan. In 2025, the prime minister stayed in office by pledging to suspend the increase in retirement age. Meanwhile, national debt has reached unprecedented levels. A reform that leaves some group worse off meets opposition through some channel: at the ballot box, in the street, or in the next government.

This paper studies what will happen after the Social Security trust fund runs out, who will be affected, and what the government can do about it. In the baseline, which I refer to as the no-reform case, the Treasury borrows to pay full benefits, as France has done in the face of political deadlock over reform, rather than letting an automatic benefit cut under the current law. Financing through borrowing debt is politically the easiest option because it defers the cost. Once national debt reaches its ceiling, benefits fall and a consumption tax rises, so the burden lands on cohorts who paid full payroll taxes over their careers, and then receive lower benefits in retirement.

I find a transition that uses only a few instruments to spread this burden across generations and leaves no birth cohort worse off, on average, than under no reform. My proposed reform is different from a complete phase-out, and leaves a smaller but positive program in the long run. The main idea of the reform is how the efficiency gains from transitioning to a smaller program can be brought forward to protect those transition cohorts, which instruments do so, and what it costs the government. Then I ask, once such a transition exists, what should be the size of the program. I compare the smaller program with a complete phase-out and show that the ranking depends on how a government weighs the generations alive today against those not yet born. I show that my proposed reform gives a higher welfare if the government care only those alive today or weights future cohorts lightly through dynastic altruism.

[1] The OASI and DI trust funds are legally separate. The combined projection assumes legislation that permits transfers between them and is the basis of our calibration. Under current law, OASI reserves alone are projected to be depleted in 2032.

The main reform also changes the macroeconomy. Without reform, aging lowers output per adult by 12% relative to 2025, and the interest rate stays at 4.6%. Under the reform, people work more and save more for their own retirement. By 2200, capital per adult is 23% higher than in the baseline, output per adult is 3% above its 2025 level. The interest rate is 4%. With public debt at 150% of GDP in both scenarios, the lower rate reduces annual interest payments by about 1% of GDP in 2200. Together with a larger tax base, this lets the consumption tax fall below the baseline rate.

Finding a welfare-improving policy reform matters because Social Security is an implicit contract between generations. Retirees made irreversible decisions based on the benefits they were promised. It will be broken immediately once the system cannot pay the full scheduled benefits, and the cost falls on the cohorts when the benefits are cut. This may even hurt the credibility of the government policy. Therefore, understanding the transitional economy is as important as understanding a long-run steady-state, especially when the policy reform is indispensable.

To answer this question, I study a large-scale overlapping-generations model in the tradition of Auerbach and Kotlikoff (1987), with the demographic and fiscal structure of Braun and Joines (2015). Households enter at 20, face mortality risk, retire at 65 and live to at most 89, and the population ages along the Trustees' projected path. Households differ in assets and skills, value the bequests they leave (De Nardi, 2004), and build skills through work (Imai and Keane, 2004; Chang et al., 2002), so the taxes that finance pensions change future earnings as well as current hours. The government runs Social Security and the Treasury as separate accounts, borrows up to a debt ceiling, and closes the remaining gap with a consumption tax, the financing margin used in Braun and Joines (2015).

I calibrate the model to match the features of the U.S. economy that matter most for the pension reform. Households save, work, and build skills at the rates observed in 2010. The economy holds three years of output in capital, adults work the hours recorded in the American Time Use Survey, and wages rise and then flatten over the working life as they do in the Current Population Survey, so that the response of saving, labor supply, and future earnings to a tax or benefit change is anchored to data. The initial wealth distribution comes from the 2010 Survey of Consumer Finances. Within each age group, the model starts with the same share of people who hold no wealth and the same inequality of wealth among those who hold some. The bequest motive is set so that estates equal 5.4% of GDP, the flow implied by wealth and mortality data, and inheritances go to each age group in the shares reported in the survey. This matters because the reform changes how much households save and how much they leave to their heirs, so the

model should start these changes from realistic levels of wealth and bequests. On the fiscal side, the model starts from the observed public debt and trust-fund reserves, Social Security's benefits in payment and benefit formula, its payroll base, and government spending from the national accounts. The population then ages along the Trustees' projection, and the model reproduces the Trustees' reserve-exhaustion year and their projected program cost in 2100.

In the baseline economy, the payroll tax stays at 12.4%. After reserves run out in 2034, the Treasury borrows to pay full benefits until debt reaches the ceiling in 2041. Then benefits fall to what payroll revenue covers, 75% of scheduled benefits at once and 63% by 2100, close to the 65% the Trustees project for 2100 (Social Security Trustees, 2026), and the consumption tax rises in the same year. Delaying the adjustment by seven years changes who pay for the shortfall. From 2034 to 2040, retirees keep full benefits, paid for by borrowing that later taxpayers service through the consumption tax. The cohorts that retire after 2041 pay the full payroll tax over their careers, receive lower pension benefits during their retirement, and pay the consumption tax as well. Any reform has to be judged by what it does for them.

This delayed-reform baseline is plausible. Conflict over who bears the cost can postpone reform (Alesina and Drazen, 1991), and holding fiscal policy fixed until debt reaches a threshold follows Hansen and İmrohoroğlu (2016). Paying full scheduled benefits after the reserves run out follows the CBO's projection convention (Congressional Budget Office, 2024). Limiting benefits to payroll revenue once borrowing stops is what current U.S. law prescribes, and other countries have done the same: Greece cut pensions once it could no longer borrow, and Sweden adjusts them to funding by rule.[2]

In the main reform, can Social Security be made smaller without any cohort losing on average relative to the baseline? The payroll tax stays at 12.4% through 2060, then falls linearly to 3.5% in 2200. Benefits follow the baseline rule: full benefits through 2040, then what payroll revenue covers, so they fall to 16% of scheduled benefits in 2200. Cohorts entering after the cut begins pay less tax and receive lower benefits. The cohorts, who pay the full 12.4% for most of their careers and retire into lower benefits, receive payroll relief and benefit supplements. The Treasury pays for both under the same 150% ceiling, so the consumption tax finances them.

I choose payroll tax reductions and benefit supplements separately for each eligible birth cohort. Payroll relief lowers the cohort's tax rate. The supplement raises its paid retirement benefit

[2]Sweden's automatic balancing mechanism lowers the indexation of pensions, without new legislation, when the system's liabilities exceed its assets; after the 2008 financial crisis it cut pensions in payment in several years. Greece financed pension deficits through public borrowing until it lost access to credit markets in 2010, after which pensions were cut repeatedly under its financial assistance programs.

by a cohort-specific percentage. I adjust both instruments using each cohort's average welfare gain relative to the baseline, measured in consumption-equivalent units. After each adjustment, I solve the full transition again, including changes in wages, interest rates, and the consumption tax needed to finance compensation. Compensation protects each birth cohort's average lifetime welfare relative to the baseline. Retirement benefits, including the supplement, can exceed the baseline level for a given cohort and year.

Each part of this reform has been used in practice. Germany, Japan, Canada, and Australia all pay part or all of their public pensions from general revenue, and so does the United States for Medicare Parts B and D (Medicare Trustees, 2025). Pension reforms are routinely phased in by birth cohort, as the 1983 Social Security amendments did with the retirement age, and Chile compensated the cohorts caught in its 1981 transition with Treasury-paid recognition bonds. Since 2022 Canada pays Old Age Security recipients aged 75 and over a benefit 10% higher than younger recipients receive. This paper combines these instruments into one transition and asks how large each must be, cohort by cohort, for no birth cohort to lose on average, and whether a government that holds its debt at a fixed ceiling can afford them after the consumption tax has adjusted.

I compare the main reform to an elimination of Social Security or the phase-out, in the spirit of McGrattan and Prescott (2017), without compensation. The payroll tax falls in equal annual steps from 12.4% in 2026 to zero in 2200. Lower payroll revenue exhausts reserves in 2033, a year earlier than in the baseline, and benefits are then limited to the program's own revenue, so current retirees face an immediate cut and benefits reach zero in 2200. The Treasury does not borrow to cover the program's shortfall, but the general budget follows the same debt rule. Debt still rises without Social Security borrowing, because the interest rate exceeds output growth and the government budget runs a primary deficit partly due to rising Medicare. Until debt reaches the ceiling in 2043, the government pays the interest by borrowing more, so interest payments grow from about 2% of GDP in 2026 to 5% in 2043.

During the phase-out, the payroll tax falls from 2026, so hours rise, and households expecting smaller benefits save more. By 2040, output per adult is 2.5% above the baseline and capital per adult is 3.6% above. Because the Treasury does not borrow to pay benefits, debt reaches the ceiling in 2043, so the consumption tax changes later. The gap then widens. In 2200, output per adult is 24% above the baseline under the phase-out and 16% under the main reform, and consumption per adult is 23% and 16% above. More capital per unit of effective labor raises the wage and lowers the interest rate, and the larger tax base lowers the consumption tax. Output, capital, hours, and consumption all end higher under the phase-out than the main reform. However, higher

aggregates do not mean that everyone gains and this is why policy implications can differ under transition. Under the main reform, consumption of those already retired barely moves as well as the consumption of working age. Under the phase-out, retirees consume 1.6% less than in the baseline already in 2026, before any benefit cut, because they expect it, and 2.4% less in 2034. Workers consume more from the start, 1% more by 2040. Both lines jump in 2041, when the baseline reduces benefits and its consumption tax rises to 9.7%.

The two policies favor different generations. The main reform is better for adults aged 38 to 88 in 2026, two thirds of today's adults, because the phase-out cuts their benefits from 2033 and leaves them little time to adjust. The phase-out is better for everyone born after 1988, who work under a lower payroll tax for longer. I use a discounted-utilitarian criterion over adults alive in 2026 and future age-20 entrants, in the spirit of Calvo and Obstfeld (1988). The phase-out's long-run gain is larger, but it comes at a cost to most current adults alive. The main reform's long-run gain is smaller, but it leaves no cohort worse off than the baseline. The main reform beats the phase-out at social discount factors β^* below about 0.87.[3] Above these rates, the phase-out yields higher aggregate welfare, even though some cohorts lose. However, once we assume that the planner has a dynastic altruism so that they care not only about own welfare but that of descendants (Becker and Barro, 1988; Córdoba, 2015), the social discount factor prefers the main reform to the phase-out. Since how much weight a later cohort should receive relative to an earlier one is a value judgment that data cannot settle (Dasgupta, 2008), I show how the welfare-ranking changes across a range of social weights.

Learning by doing is essential for model result. Without it, a lower payroll tax no longer raises skills, so the long-run gain falls from 7.7% to 2.8% of consumption. The efficiency gains are then too small and come too late to pay for protecting every cohort. The bequest motive is not essential. Without it, the reform still leaves no cohort worse off, whether the assets of the deceased go to survivors of the same cohort, as in an annuity, or are passed on as inheritances. How the compensation is financed matters little. Paying for the payroll relief with an extra payroll tax that only cohorts born in 2044 or later pay, instead of from general revenue, lowers social welfare slightly, from 0.17% to 0.16%. How fast the payroll tax is reduced matters more. Spreading the same cut over a longer period, so that the rate reaches 3.5% in 2300 rather than 2200, still protects every cohort but lowers social welfare to 0.10%, because cohorts born after 2045 pay the higher tax for longer and receive their gains later. Finally, the no-loss result holds for cohort

[3]This pure annual rate 14.5% is above the rates in the literature of environmental economics and climate change, from the low pure rates in Stern (2007) to the 5.8% market-based calibration of van den Bremer and van der Ploeg (2021).

averages, not for every household. In the main reform, 4.3% of the adults alive in 2026 lose, by at most 0.17% of consumption. Sizing the same two instruments to protect the worst-off household in each cohort protects everyone and raises the present-value cost from 2.2% to 3.3% of 2026 GDP.

Related Literature. The paper belongs to the literature on the transition cost of pension reform. Breyer (1989), Geanakoplos et al. (1998), and Sinn (2000) show that the windfall paid to the first generation of a pay-as-you-go system is an implicit debt. Moving to a funded system, in which each generation saves for its own pensions instead of being paid by the next generation's taxes, cannot erase this debt, so some generation must carry it. Quantitative life-cycle models confirm the pattern along the transition (Auerbach and Kotlikoff, 1987; Huang et al., 1997; Conesa and Krueger, 1999; Nishiyama and Smetters, 2007). Others compare ways to finance the shortfall in an aging economy with public debt (Kitao, 2014; Braun and Joines, 2015; İmrohoroğlu et al., 2016). McGrattan and Prescott (2017) is the closest to this paper: in an economy with annuities and no endogenous human capital, they find a phase-out of Social Security and Medicare that leaves every cohort better off, but together with a broader income-tax reform. I study a transition of Social Security system that the US would most likely to face once the trust fund runs out in next few years. Such economy has not been studied before since most papers in the literature assume a policy reform with no exhaustion in the trust fund. Once the trust fund runs out, Social Security will face an abrupt deficit of $\$0.3$ trillion, which could be financed by higher payroll taxes, lower benefits, or additional government borrowing from national debt.

A second literature asks what size of Social Security is optimal. Positive answers rest on insurance: against longevity when annuity markets are missing, against earnings and health risk, and through redistribution across different types (İmrohoroğlu et al., 1995; Hubbard et al., 1995; Huggett and Ventura, 1999; Conesa and Krueger, 1999; Hosseini and Shourideh, 2019), with Conesa and Garriga (2008) and Peterman and Sager (2022) studying the optimal mix of social security, taxes, and public debt in life-cycle economies. Jang et al. (2025) show, with the steady-state welfare of a newborn behind the veil of ignorance as the criterion, that once the policy design is changed, a larger size is optimal. This paper studies the transition instead. Along the transition, the size of the program becomes an intergenerational choice: it decides which cohorts gain and which lose, and the answer depends on how much weight society places on each cohort. I show that the smaller program yields higher social welfare than the phase-out when a planner gives sufficient weight to the cohorts that bear the transition costs or weights future cohorts lightly through dynastic altruism.

Lastly, two features of the model have not been studied together, and the question needs both. Learning by doing (Imai and Keane, 2004; Chang et al., 2002; Wallenius, 2011) decides how large the efficiency gains from a lower payroll tax are and when they arrive: more work raises skills and future earnings, so the tax base grows by more, and later, than the hours response alone would deliver. Blandin (2021) and Peterman (2016) show that this changes the welfare effects of payroll and income taxes. The debt ceiling, enforced by a consumption tax (Bohn, 1998; Altig et al., 2001; Hansen and İmrohoroğlu, 2016; İmrohoroğlu et al., 2016), decides whether those gains can be used to protect the transition. The debt ceiling is a percentage of GDP, so the government cannot borrow more to pay for compensation. It can borrow more only if the reform raises output, so the reform has to pay for the compensation out of its own gains. The money is raised through the consumption tax, which retirees also pay, so the people being compensated cover part of the cost themselves. The literature has studied the two apart, skill formation in models of tax reform without a constraint on public borrowing and debt ceilings in aging economies without skill formation. Whether a transition can be financed without losers depends on both. Without the debt ceiling, any transition could be made welfare-improving by borrowing against future output. Without learning by doing, the efficiency gains are not sufficient.

2 Model

The model is a general-equilibrium overlapping-generations economy based on Auerbach and Kotlikoff (1987). Its demographic and fiscal components are similar to those in Braun and Joines (2015). Each model period is one year.

Households enter the model at age 20. They retire at age 65 and die by age 90. Households differ in their assets and skills. They choose consumption, saving, and hours of work. They cannot borrow. Their skills increase through work. Households also value the bequests they leave. Mortality is their only source of idiosyncratic risk. Firms use capital and efficiency units of labor to produce one good. The government has four components. Social Security pays benefits from payroll taxes and its trust fund. Medicare and a social safety net provide support for the elderly. The Treasury uses a consumption tax to balance the overall government budget. Households have perfect foresight along deterministic policy paths.

2.1 Demographics and productivity

Let $N_{j,t}$ be the population of age j at the start of year t, and $\pi_{j,t}$ its probability of surviving that year. Within each calendar year, people of the same age face the same survival probability. Both series use historical estimates and the intermediate projection of the 2026 Trustees Report through 2100 (Appendix Figure A.2). After 2100, I use the Markov population process of Braun and Joines (2015) to extend the population to a stationary distribution.

I measure population relative to $\overline{N}^W$, the number of adults aged 20–64 in 2010:

$$n_{j,t} = \frac{N_{j,t}}{\overline{N}^W}, \qquad n_t^W = \sum_{j=20}^{64} n_{j,t}, \tag{1}$$

where n_t^W is the working-age population in these units.

The old-age dependency ratio is the number of adults aged 65–89 per 100 adults aged 20–64. This ratio approximately doubles between 2010 and the stationary population (Figure A.1). Labor-augmenting productivity A_t grows at a constant rate g. I normalize it so that $A_{2010} = 1$. I express quantities in 2010 dollars. For computation, I divide quantities by A_t so that this gives a balanced-growth path in the long run.

2.2 Environment

A household of age j in year t starts the year with assets $a \geq 0$ and skill $h > 0$. It chooses consumption c, saving $x \geq 0$, and hours of work $\ell \in [0, 1)$. Hours are measured as a fraction of sixteen hours per day. Households must retire at age 65, so $\ell = 0$ from that age. Households adjust labor supply only through hours of work before age 65. Following Braun and Joines (2015), households face no idiosyncratic earnings risk.[4]

Households value consumption and leisure through a constant-relative-risk-aversion function of a Cobb–Douglas composite of the two:

$$u(c, \ell) = \frac{[c(1-\ell)^v]^{1-\sigma} - 1}{1-\sigma}$$

[4] I exclude individual earnings shocks to focus on how saving, work, and skill accumulation respond to benefit rules and their financing. This assumption also makes repeated transition and compensation calculations manageable. Assets and skills are the only household state variables. Earnings risk would give households a precautionary saving motive (Conesa and Krueger, 1999; Nishiyama and Smetters, 2007). It could also change their ability to absorb benefit cuts. Thus, the quantitative valuations depend on this assumption.

and the estate x they leave at death:

$$\mathcal{W}(x) = \vartheta \, \frac{(1 + x/b_*)^{1-\sigma} - 1}{1 - \sigma}.$$

The bequest term $\mathcal{W}$ represents a warm-glow motive, as in De Nardi (2004). Households receive utility from the estate they leave. The scale b_* is set in the calibration. Bequests are a luxury good in the model. Households below a wealth threshold leave no estate. Most estates come from wealthy households, as in the data. In year $t+1$, the value of a bequest x is $\mathcal{W}_{t+1}(x) = A_{t+1}^{1-\sigma}\mathcal{W}(x/A_{t+1})$. This adjustment makes the bequest motive consistent with balanced growth.

Skills increase through learning by doing, as in Imai and Keane (2004) and Chang et al. (2002):

$$h_{j+1,t+1} = (1 - \delta_h)h_{j,t} + \eta \, \ell_{j,t} h_{j,t}^{\gamma} \tag{2}$$

where the parameter δ_h is the rate of skill depreciation. Work produces current earnings. It also builds skills that increase future earning capacity. The parameter η measures the efficiency of learning by doing. The parameter γ determines how the skill gain depends on existing skills. The marginal effect of work time on next year's skill is $\partial h_{j+1,t+1}/\partial \ell_{j,t} = \eta h_{j,t}^{\gamma}$. Younger workers have more years before retirement to earn a return on this additional skill. Learning by doing links current work incentives to future earnings. A household can work more to prepare for a benefit cut, building experience as well as current income. Higher benefits and the taxes used to finance them can reduce hours, skills, and future earnings. The model captures these responses through the hours choice.

For a household of age j in year t with assets $a_{j,t}$ and skill $h_{j,t}$, the budget constraint is

$$\begin{aligned} \underbrace{(1+\tau_t^c)c_{j,t}}_{\text{consumption}} + \underbrace{x_{j,t}}_{\text{saving}} + \underbrace{o_{j,t}}_{\substack{\text{medical}\\ \text{bills}}} = {} & \underbrace{[1 + (1-\tau^k)r_t](a_{j,t} + e_{j,t})}_{\text{assets and inheritance, after tax}} \\ & + \underbrace{(1 - \tau_t^{ss} - \tau^G) w_t h_{j,t} \ell_{j,t}}_{\text{after-tax earnings}} \\ & + \underbrace{b_{j,t} + s_{j,t}}_{\substack{\text{Social Security}\\ \text{and other transfers}}} + \underbrace{f_{j,t}(a_{j,t})}_{\substack{\text{floor}\\ \text{payment}}}, \end{aligned} \tag{3}$$

where $o_{j,t}$ is the out-of-pocket share of age-specific medical spending. The term $e_{j,t}$ is the inheritance received. The wage per efficiency unit is w_t. The Social Security benefit is $b_{j,t}$.[5] Other

[5]Benefit amounts in the model are net of the federal income tax on benefits. This tax revenue is credited to the

transfers are $s_{j,t}$, and the consumption-floor payment is $f_{j,t}(a)$.

Assets, skill, and choices differ across households within an age group. Medical bills, inheritances, benefits, and other cash transfers are the same for everyone in an age group in a given year. Floor payments also depend on household assets. The consumption tax rate is τ_t^c. The Social Security payroll tax rate is τ_t^{ss}. The earnings tax rate is τ^G, and the capital-income tax rate is τ^k. The earnings and capital-income tax rates remain at their 2010 values. The consumption tax adjusts when the Treasury must balance its budget. From age 65, the government guarantees consumption of at least $\underline{c}_t$. If a retiree's resources are less than $(1+\tau_t^c)\underline{c}_t$, the retiree receives the difference as the floor payment $f_{j,t}(a)$. The retiree consumes $\underline{c}_t$ and saves nothing. While the retiree receives support, each additional dollar of resources reduces the floor payment by one dollar. This means-tested consumption guarantee follows Hubbard et al. (1995) and De Nardi et al. (2010).

Medicare and medical spending. Medical spending is exogenous and deterministic: it varies by age but not across households within an age group. Let m_j be health-care spending per person at age j in 2010. Medicare pays a share κ_j in kind, so it enters government spending Z_t and not household utility. Households pay an out-of-pocket share ω_j, the bills $o_{j,t}$ in the budget constraint. A common factor ζ_t scales the age profile each year so that Medicare outlays equal an exogenous share $\mathcal{M}_t$ of output:

$$\sum_j n_{j,t}\,\kappa_j m_j\,\zeta_t = \mathcal{M}_t Y_t, \qquad o_{j,t} = \omega_j m_j\,\zeta_t. \tag{4}$$

Out-of-pocket bills rise steeply with age, which gives households a reason to save for old age. Because both are tied to Y_t, a policy that raises output raises Medicare outlays and medical bills in proportion.

Recursive Formulation A household of age j in year t enters with assets a and skill h and chooses consumption c, saving x, and hours ℓ. With probability $\pi_{j,t}$ it survives to age $j+1$: its saving becomes next year's assets, $a_{j+1,t+1} = \kappa_{j,t}x_{j,t}$, where $\kappa_{j,t}$ is the factor defined in Appendix A.1, and its skill follows (2). With probability $1-\pi_{j,t}$ individual dies and its saving is the bequest he leaves, valued by $\mathcal{W}_{t+1}$. Its value function satisfies

$$V_{j,t}(a,h) = \max_{c,x,\ell}\Big\{u(c,\ell) + \beta\big[\pi_{j,t}V_{j+1,t+1}(\kappa_{j,t}x, h_{j+1,t+1}) + (1-\pi_{j,t})\mathcal{W}_{t+1}(x)\big]\Big\}, \tag{5}$$

trust fund, so it cancels out of the fund's accounts. Appendix A.2 gives the rate used.

subject to the budget constraint (3) and the consumption floor, with $\ell = 0$ from age 65. Survival ends at 89, $\pi_{89,t} = 0$, so the last period's saving is left entirely as an estate.

Three features of the solution affect the results. First, the household compares the leisure cost of an additional hour of work with its wage return and skill return. An earnings tax reduces both returns to work. Second, if next year's consumption floor binds, the marginal value of saving is zero. Third, skills and work are complementary. More skill increases the return to work. Work also builds additional skill. With the consumption floor, these effects mean that the lifetime problem need not be concave.

2.3 Estates and firms

Assets left by households that die during the year are collected and distributed the next year as inheritances. This rule follows the distribution of accidental bequests in Huggett (1996) and Braun and Joines (2015). Each age group receives a share ς_j of these estates, with $\sum_j \varsigma_j = 1$. This share is divided equally among the members of that age group. The shares differ across ages and remain fixed over time. They are based on the ages at which people received inheritances in the 2010 Survey of Consumer Finances:

$$\mathcal{E}_{t+1} = \sum_j n_{j,t}(1 - \pi_{j,t}) \int x_{j,t}\, d\mu_{j,t}, \qquad e_{j,t} = \varsigma_j \mathcal{E}_t / n_{j,t}, \tag{6}$$

where $\mu_{j,t}$ is the distribution of assets a and skill h among households of age j.

Competitive firms use capital K_t and efficiency units of labor L_t to produce output: $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$. Factor prices equal marginal products. The wage is $w_t = (1-\alpha)Y_t/L_t$. The net return on capital is $r_t = \alpha Y_t/K_t - \delta$, where δ is the depreciation rate.

2.4 Social Security

The model represents the combined OASDI program with one trust fund and follows the Trustees' projections (Social Security Trustees, 2026). Workers pay a 12.4% payroll tax on covered earnings up to the taxable maximum. The model has neither a taxable maximum nor uncovered work, so it applies the effective rate $\tau_t^{ss} = 0.124\,\zeta_t$ to all earnings, where ζ_t is the projected share of labor income subject to the tax. Payroll receipts are $\mathcal{P}_t = \tau_t^{ss} w_t L_t$.[6]

A retiree's benefit is set in the year the cohort turns 65 as a fixed fraction of average earnings

[6] Calligraphic letters denote program aggregates and lowercase letters amounts per person.

in that year, and it stays constant in real terms for life. It depends on the cohort's average earnings, not on the retiree's own earnings history. This assumption was to make model tractable. Let $b^R_{j,t}$ be the scheduled retired-worker benefit per member of age group j in year t, and $b^O_{j,t}$ the other OASDI cash benefits per member: disability, spouse, child, and survivor benefits, which follow their projected share of scheduled cost. Scheduled spending is

$$\mathcal{H}_t = \sum_j n_{j,t}\big(b^R_{j,t} + b^O_{j,t}\big). \tag{7}$$

Appendix A.5 gives the benefit formula, the adjustment for the rise in the normal retirement age, coverage rates, the income tax on benefits, the benefits already in payment in 2010, and the paths of the taxable share and other benefits.

The trust fund holds reserves $\mathcal{F}_t$ in Treasury securities with a special gross return R^F_t. It receives payroll receipts and Treasury transfers $\mathcal{J}_t$, pays other program expenses $\mathcal{O}_t$, and cannot borrow. It pays only the fraction ϕ_t of scheduled benefits that its resources cover, the paid share:

$$\phi_t = \min\left\{1, \frac{R^F_t \mathcal{F}_t + \mathcal{P}_t + \mathcal{J}_t - \mathcal{O}_t}{\mathcal{H}_t}\right\}, \tag{8}$$

$$\mathcal{F}_{t+1} = R^F_t \mathcal{F}_t + \mathcal{P}_t + \mathcal{J}_t - \phi_t \mathcal{H}_t - \mathcal{O}_t \ \geq 0. \tag{9}$$

Each member of age group j receives $b_{j,t} = \chi_{j,t}\,\phi_t(b^R_{j,t} + b^O_{j,t})$, where $\chi_{j,t} \geq 1$ is a supplement factor by birth year that equals one unless a reform sets it.

Policy instruments A policy is a path for the payroll rate τ^{ss}_t, a rule for the Treasury transfers $\mathcal{J}_t$, and, for a reform, two cohort-specific instruments. In the baseline the payroll rate stays at 12.4% and the Treasury transfers whatever the fund needs to pay scheduled benefits, as long as it can borrow within its debt limit. Once the limit binds, the transfers shrink to the statutory reimbursements, the fund pays only what its own resources cover, and the paid share falls below one, $\phi_t < 1$. I call these funding-linked benefit cuts.[7] A reform lowers the payroll rate along an announced path and may protect chosen birth cohorts with two instruments: a payroll relief

[7] Sweden's automatic balancing mechanism, introduced with the 1999 pension reform, is the closest rule-based counterpart. Pensions in payment are normally indexed to average wage growth. When the system's liabilities exceed its assets, indexation switches to a lower balancing index until funding is restored, and each forward as a lower benefit level. As in the baseline, the adjustment falls on benefits rather than contribution rates, is triggered by insufficient funding, and lands on current retirees. It differs in two respects: the trigger is the pension system's own balance sheet not on total public debt, and once balance is restored. Pensions grow faster than under normal indexation until the lost indexation is recovered, whereas here benefits remain at what the fund's resources cover.

$\upsilon_{j,t} \geq 0$, so that a member of age group j pays the rate $\tau_t^{ss} - \upsilon_{j,t}$ on its earnings, and a benefit supplement $\chi_{j,t} > 1$, which raises the benefit the member receives. The Treasury pays for both: the revenue the relief gives up and the cost of the supplement are added to $\mathcal{J}_t$. Section 4 describes how the two instruments are set in more detail.

2.5 Government spending and the Treasury

Government spending outside Social Security, Z_t, has four parts. Purchases are federal consumption and investment, defense and nondefense, net of depreciation. Cash transfers, the $s_{j,t}$ in the household budget, are the federal transfers other than Social Security and Medicare: unemployment insurance, workers' compensation, nutrition assistance, refundable tax credits, and the remaining federal transfer programs. Medicare is supplied in kind as a share $\mathcal{M}_t$ of output that follows the projected path (Attanasio et al., 2011; Braun and Joines, 2015). Consumption-floor payments are the fourth part. Purchases are a fixed 2010 real amount per person of any age, and each transfer a fixed 2010 amount per person in its recipient group, adults 65 and over or adults of working age. Spending outside Social Security are imposed exogenously. Households pay the out-of-pocket part of medical spending themselves, the age-specific bills $o_{j,t}$ in the budget constraint.

The Treasury collects income-tax revenue

$$T_t = \tau^G w_t L_t + \tau^k r_t (K_t + B_t).$$

It also collects consumption-tax revenue $\tau_t^c C_t$. After Social Security is combined with the rest of the government, debt evolves as

$$B_{t+1} = (1 + r_t) B_t + \underbrace{\phi_t \mathcal{H}_t + \mathcal{O}_t - \mathcal{P}_t}_{\text{Social Security deficit}} + \underbrace{Z_t - T_t - \tau_t^c C_t}_{\substack{\text{non-Social-Security} \\ \text{primary deficit}}}. \tag{10}$$

The Social Security deficit equals benefits plus other program costs, less payroll receipts. Treasury transfers to the fund cancel on consolidation. The primary deficit of the rest of the government equals purchases, transfers, Medicare, and consumption-floor payments, less income-tax and consumption-tax revenue.

The consumption tax limits next year's debt to $\bar{b} Y_{t+1}$. If debt remain within the limit without

this tax, the tax rate is zero. Otherwise, the tax raises the revenue needed to respect the ceiling:

$$\tau_t^c = \max\left\{0, \frac{\widetilde{B}_{t+1} - \bar{b}Y_{t+1}}{C_t}\right\}, \tag{11}$$

where $\widetilde{B}_{t+1}$ is next year's debt with a zero consumption tax. The limit is $\bar{b} = 1$, or 100% of GDP, in the calibration, and $\bar{b} = 1.5$ in the policy experiments. It is an upper bound, not a requirement to hold debt at the ceiling. The government can accumulate assets when revenue exceeds spending and the consumption tax is zero. Lower benefits can therefore lower taxes, repay debt, or build government assets. This fiscal rule follows a debt target in the spirit of Bohn (1998) where the consumption tax enforces the target. By contrast, Kitao (2014) holds debt per person fixed and adjusts the labor-income tax to balance the general budget.

In the model, the government uses a consumption tax to finance budget shortfalls when further borrowing would exceed the debt limit. These shortfalls can arise from aging, Medicare spending, or Treasury transfers to Social Security. This tax is a standard choice in models of aging and government debt (Altig et al., 2001; Braun and Joines, 2015; Hansen and İmrohoroğlu, 2016; İmrohoroğlu et al., 2016). It distorts workers' incentives to work, but retirees also pay the tax. Thus, retirees bear part of the cost of their own benefits.[8]

2.6 Equilibrium

The equilibrium concept is a competitive equilibrium with perfect foresight along a transition path, as in Auerbach and Kotlikoff (1987), Ríos-Rull (1996), and Huggett (1996). Households and firms take prices and policy rules as given, and the price and aggregate paths they expect are the ones their choices produce. Tax revenue is whatever the resulting earnings, capital income, and consumption yield, and the government's budget and financing rules hold in every year.

Definition 1 (Perfect-foresight Competitive Equilibrium). *Let $t_0 = 2010$. Take as given preferences, technology, the demographic paths $\{n_{j,t}, \pi_{j,t}\}$, productivity growth g, the benefit, coverage, spending, Medicare, and consumption-floor rules, the income-tax rates (τ^G, τ^k), the payroll-tax path τ_t^{ss}, and the debt ceiling $\bar{b}$. Initial conditions are K_{t_0}, B_{t_0}, $\mathcal{F}_{t_0}$, benefits already in payment, $\mathcal{E}_{t_0} = 0$, and distributions $\{\mu_{j,t_0}\}_{j=20}^{89}$ with $\sum_j n_{j,t_0} \int a \, d\mu_{j,t_0} = K_{t_0} + B_{t_0}$.*

[8] Nishiyama and Smetters (2005) show that replacing a progressive income tax with a flat consumption tax can reduce efficiency when households cannot insure against wage shocks. That channel is absent here. The model has no individual earnings risk. It also adds the consumption tax to existing income taxes instead of replacing them.

A perfect-foresight competitive equilibrium is a sequence for $t \geq t_0$ of household policies and values $\{c_{j,t}, x_{j,t}, \ell_{j,t}, V_{j,t}\}$, distributions $\{\mu_{j,t}\}$, prices $\{w_t, r_t\}$, aggregates $\{K_t, L_t, Y_t, I_t, C_t, \mathcal{E}_t\}$, and fiscal outcomes $\{\tau_t^c, \phi_t, B_t, \mathcal{F}_t, Z_t\}$ such that:

1. Households optimize. *$V_{j,t}$ and the policies solve* (5) *subject to* (2), (3), *the consumption floor, $x \geq 0$, and $\ell_{j,t} = 0$ for $j \geq 65$, at the equilibrium prices, taxes, transfers, and inheritances.*

2. Distributions are consistent with household choices. *For $20 \leq j < 89$ and any set of states $\mathcal{D}$,*

$$\mu_{j+1,t+1}(\mathcal{D}) = \int \mathbf{1}\{\mathbf{T}_{j,t}(\mathbf{s}) \in \mathcal{D}\}\, d\mu_{j,t}(\mathbf{s}),$$

   where $\mathbf{T}_{j,t}(\mathbf{s}) = (a',\, h')$ is the state that a household with state $\mathbf{s} = (a, h)$ reaches at age $j+1$ in year $t+1$, with h' given by (2) *and $\mu_{20,t}$ places entrants at zero assets with the initial skill distribution.*

3. Firms optimize and aggregates are consistent. $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$, $w_t = (1-\alpha)Y_t/L_t$, $r_t = \alpha Y_t/K_t - \delta$, *and*

$$L_t = \sum_{j=20}^{64} n_{j,t} \int h\ell_{j,t}\, d\mu_{j,t}, \quad C_t = \sum_{j=20}^{89} n_{j,t} \int c_{j,t}\, d\mu_{j,t},$$

   with estates given by (6).

4. Government rules hold. *The trust fund follows* (8)–(9), *Treasury transfers $\mathcal{J}_t$ follow the policy's rule, the Treasury satisfies* (10), *and the consumption tax satisfies* (11), *equivalently*

$$\tau_t^c \geq 0, \qquad B_{t+1} \leq \bar{b} Y_{t+1}, \qquad \tau_t^c \big(\bar{b} Y_{t+1} - B_{t+1}\big) = 0. \tag{12}$$

5. Markets clear.

$$K_{t+1} + B_{t+1} = \sum_{j=20}^{89} n_{j,t} \int x_{j,t}\, d\mu_{j,t}, \tag{13}$$

   so that, by item 2, the factor $\kappa_{j,t}$, and (6), *household assets plus estates at the start of each year equal $K_t + B_t$. With $I_t = K_{t+1} - (1-\delta)K_t$,*

$$Y_t = C_t + I_t + G_t, \tag{14}$$

   where G_t is all other spending on goods and services: government purchases, Medicare, out-of-pocket medical spending, and Social Security administration.

Three remarks. First, Social Security reserves are claims between government accounts. Thus, they do not appear in (13). Treasury reimbursements to the fund cancel when the government accounts are consolidated. Second, a cash transfer gives the household money to spend or save. The transfer itself does not use goods or services. Consumption financed by the transfer is already included in C_t. Adding cash transfers to (14) would double-count this consumption. The goods-market equation follows from combining the household and government budgets with asset-market clearing. Third, the equilibrium determines when Social Security reserves are exhausted, when the debt ceiling binds, and the long-run paid share of scheduled benefits.

Definition 2 (Balanced-growth equilibrium)**.** *A balanced-growth equilibrium is a perfect-foresight competitive equilibrium with stationary demographic inputs. Within each age group, the distribution of $(a/A_t, h)$ is constant. Hours, skills, the interest rate, tax rates, and ϕ_t are also constant. Output, capital, consumption, wages, benefits, and fiscal stocks grow at rate g.*

The transition equilibrium is a fixed point in the space of aggregate paths. I fix a horizon of 750 years after 2010 and take the balanced-growth equilibrium of Definition 2 as the terminal state. A guess consists of the paths of capital K_t, effective labor L_t, estates $\mathcal{E}_t$, consumption C_t, and consumption-floor over the horizon. Given the guess, prices follow from the firm's conditions, and the fiscal rules deliver the paid share, the consumption tax, debt, and reserves year by year, the consumption tax being computed on the guessed C_t. The household problem is then solved backward from the terminal value functions and the distributions forward from the 2010 initial conditions, and aggregating the resulting choices gives implied paths of the same five objects. An equilibrium is a guess that reproduces itself. A policy announced in 2026 is an unanticipated change: the path through 2025 is the baseline's, every household alive in 2026 starts from its baseline state in that year, and the fixed point is computed from 2026 onward.

3 Calibration

I calibrate the model to the U.S. economy in 2010. Household parameters are chosen in the balanced-growth equilibrium of Definition 2 with 2010 mortality rates and fiscal rules, which I call the 2010 stationary equilibrium. Quantities are detrended by productivity growth, and the trust fund has no reserves in it.

The *baseline* is the transition that determines the state inherited by fiscal reforms in 2026. It starts from the 2010 population, public debt, and trust-fund reserves, together with the calibrated capital stock and household distribution. Projected demographics and program rules then

determine the path to 2026 and beyond. Appendix A gives the sources and construction of the calibration inputs.

3.1 Externally assigned parameters

Table A.1 lists the parameters set outside the model and their sources. The capital share is $\alpha = 1/3$, which matches a labor share of two thirds in the national accounts. Productivity grows at $g = 1.13\%$ a year, the real-wage growth assumption of the Trustees' intermediate projection, so that the model and the projection it is compared with share the growth rate that drives payroll and benefits. Depreciation is 6.61% a year. Given the capital share, this is the rate at which a capital–output ratio of 3, the household calibration target, delivers a pre-tax return of 4.5% and an after-tax return near 3%, in line with observed returns on wealth. Utility curvature is $\sigma = 2$, an intertemporal elasticity of substitution of one half. Earnings and health are deterministic, so σ governs how willing households are to shift consumption across years rather than attitudes to risk, and the value is in the range life-cycle models of retirement. The bequest scale b_* is the average retired-worker benefit in payment in 2010. Setting the threshold at a retiree's annual benefit makes estates a luxury good at the scale of De Nardi (2004), and the strength of the motive is fitted below. The skill exponent $\gamma = 0.05$ is taken from Fan et al. (2024). It makes the skill gain from an hour of work nearly proportional to hours, and it is held fixed when learning efficiency and skill depreciation are fitted, because the two age–wage ratios in the calibration cannot identify a third skill parameter. The capital-income tax rate is 30%, the effective rate in Kitao (2014) that combines corporate and personal taxes on capital income. Appendix A.2 gives the construction of each value.

3.2 Joint calibration to 2010 moments

Five household parameters are fitted jointly to five 2010 moments: the discount factor to the capital–output ratio, the leisure weight to average hours, learning efficiency and skill depreciation to two age–wage ratios, and bequest strength to the flow of estates relative to GDP. At each trial parameter vector I solve the household problem and the stationary distribution, so every moment is an equilibrium outcome. Table 1 reports the fitted values. Each parameter is listed with the moment it mainly governs, although all five are determined together.

The capital–output ratio of 3 pins down patience. The primitive discount factor is $\beta = 1.0099$, which after the adjustment for productivity growth becomes $\widetilde{\beta} = \beta(1+g)^{1-\sigma} = 0.9986$. This is

Table 1: Jointly calibrated household parameters

Parameter		Value	Principal target	Source	Moment	
					Data	Model
Discount factor	$\widetilde{\beta}$	0.9986	Capital–output ratio	Literature	3.00	3.00
Leisure weight	v	3.820	Hours per day, out of 16	ATUS	3.93	3.93
Learning coefficient	η	0.4507	Wage ratio, ages 40–44 / 20–24	CPS-ORG	1.905	1.905
Skill depreciation	δ_h	0.0471	Wage ratio, ages 60–64 / 40–44	CPS-ORG	1.007	1.007
Bequest strength	ϑ	150.3	Estates / GDP (%)	SCF, SSA	5.38	5.38

Notes: The five parameters are fitted jointly. Rows associate each parameter with its principal economic role. Literature: De Nardi et al. (1999) and Cagetti and De Nardi (2006). ATUS: mean work hours per calendar day in the 2010 American Time Use Survey, for all adults aged 20–64. The mean includes nonworkers and zero-work days. The pooled mean is the hours target. CPS-ORG: 2010 Current Population Survey outgoing rotation groups, workers in the indicated age groups. SCF, SSA: Survey of Consumer Finances wealth and Social Security mortality data. The model estate moment uses the 2010 stationary equilibrium's age profiles, weighted by the 2010 population and scaled to initial private wealth. Appendix A gives sample definitions and moment construction.

the factor that governs private saving. It is distinct from the social discount factor β^* that weights future cohorts in the policy comparisons.

The hours target sets the level of labor supply. Adults aged 20–64 in the 2010 American Time Use Survey work 3.93 hours per calendar day on average, counting nonworkers and days without work. The leisure weight matches this pooled mean in the 2010 stationary equilibrium, and hours at each age are model outcomes.

The estate target sets the strength of the bequest motive. I combine SCF wealth by age with SSA death probabilities, scale wealth to capital plus public debt, and obtain an annual flow of estates of 5.38% of GDP. The model counterpart uses the stationary equilibrium's age profiles with the same 2010 population weights and the same scaling.

Human capital and the age–wage profile

The model has no exogenous age-productivity schedule. Wages rise and flatten over the working life only because work builds skill under (2) and depreciation erodes it, so the observed age–wage profile is what disciplines the skill technology. In proportional terms,

$$\frac{h_{j+1} - h_j}{h_j} = \eta \ell_j h_j^{\gamma - 1} - \delta_h, \tag{15}$$

so the learning efficiency η scales the gain from an hour of work and the depreciation rate δ_h the annual loss, with the percentage gain falling as skill accumulates. Average skill at entry is normalized to one.

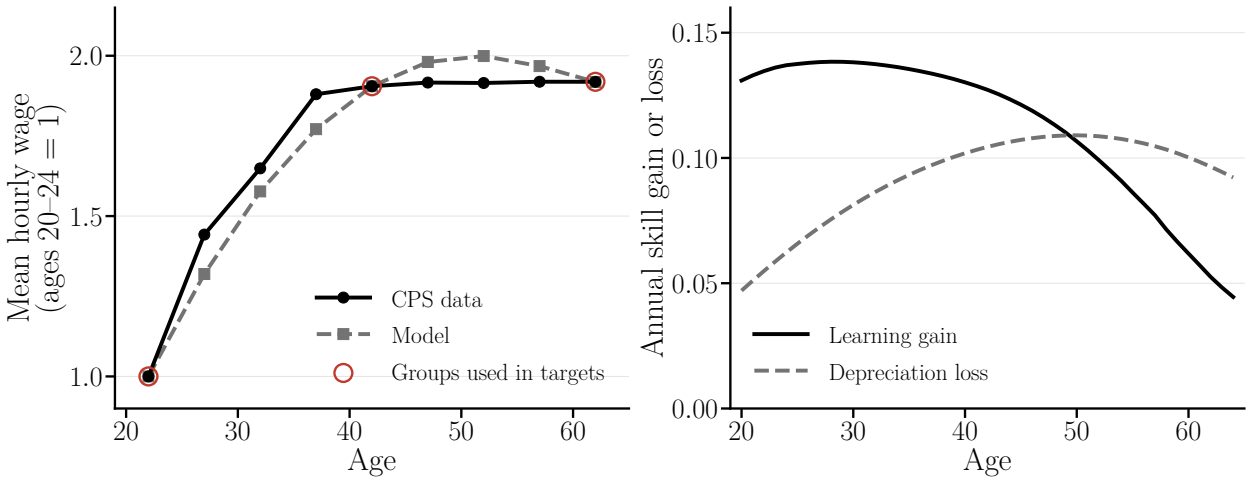


Figure 1: Human capital calibration and the age–wage profile

Notes. Left: mean hourly wages in the 2010 CPS outgoing rotation groups (earnings weights) and the 2010 stationary equilibrium (population weights, households with positive hours), each divided by its own mean at ages 20–24. Points are five-year age groups at their midpoints; red circles mark the two calibration targets. Right: mean skill gained through work and lost through depreciation at each age, over all households; the difference is the change in mean skill. Mean entrant skill is one.

Two wage ratios from the 2010 CPS outgoing rotation groups identify the two parameters: the ratio of mean hourly wages at ages 40–44 to ages 20–24, 1.905, which captures the rise, and the ratio at ages 60–64 to ages 40–44, 1.007, which captures the plateau.[9] Model wage means apply the same positive-hours restriction as the CPS sample and use the stationary population weights, so that hours affect wages only through skill and through who works. Taking ratios removes the wage per unit of skill, allowing me to identify the parameters. The fitted values are $\eta = 0.4507$ and $\delta_h = 0.0471$.

Figure 1 (Appendix Table A.2) compares all nine age groups, each profile divided by its mean at ages 20–24. The model matches the two targets, 1.905 and 1.007, understates wage growth at ages 25–39, and slightly overstates wages at ages 45–59. The largest gap is 8.5% at ages 25–29. The right panel shows why the profile has its shape. Learning exceeds depreciation early in life. As skill accumulates and hours fall, the gain shrinks relative to the loss, the two cross near age 50, where mean skill peaks, and depreciation dominates thereafter. Households that stop working leave the wage sample but stay in mean skill, which moderates the measured wage decline.

[9]The targets are cross-sectional means. The calibration does not estimate separate cohort effects or target wage dispersion at older ages. Skill paths are deterministic conditional on initial skill and household choices.

3.3 Initial conditions in 2010

The baseline starts in 2010 from observed aggregates and a constructed cross-section of households. Population and survival probabilities follow the Trustees' projections, with population measured in units of 187 million, the number of adults aged 20–64 in 2010. The old-age dependency ratio rises from 20.7 in 2010 to 45.3 in the terminal population reached in 2190, which is the demographic pressure the baseline carries. Medicare and out-of-pocket medical spending by age come from the National Health Expenditure Accounts, and future Medicare spending follows the 2025 Medicare Trustees projection. Initial capital is three times GDP, the calibration target, public debt and trust-fund reserves take their observed 2010 values, and undistributed estates start at zero. Table 2 reports these stocks together with their 2026 values.

Table 2: Initial stocks and the inherited reform state

	2010	2026
Capital / GDP	3.00	3.07
Public debt / GDP (%)	52.98	66.44
Trust-fund reserves / GDP (%)	17.23	8.74
Undistributed estates / GDP (%)	0.00	6.18

Notes: Stocks are measured at the beginning of the year. Initial capital is set to the calibration target. Initial public debt and trust-fund reserves are observed, and undistributed estates start at zero. Mean assets and skills by age follow the 2010 stationary equilibrium. Assets are scaled so that total household wealth equals capital plus public debt. The 2026 column reports the baseline state inherited by all reforms.

Household heterogeneity

Households share preferences and differ in resources. New cohorts enter at age 20 with zero assets and skills drawn from a lognormal distribution around the mean, with a log standard deviation of 0.493, the dispersion of CPS wages among workers aged 20–24. Skill differences then produce differences in wages, hours, and saving, and wealth and skill evolve jointly through household choices. The cohorts already alive in 2010 receive the same skill dispersion around the 2010 stationary equilibrium's mean skill at their age, and in addition differ in wealth. Assets and skills are drawn independently within each age. Appendix A.4 gives the numerical construction.

The SCF sets the shape of the wealth distribution within each five-year age group: a mass at zero matching the observed zero-asset share and a lognormal positive part matching the standard deviation of log positive wealth. Zero-asset shares are 30.0% at ages 20–24 and 5.1% at ages 65–69, and the log standard deviations range from 1.67 to 2.02. The level comes from one scale factor

applied to the stationary equilibrium's mean asset profile, chosen so that total assets at the 2010 population weights equal capital plus public debt, 3.530 times GDP. The scaling preserves the age profile and the relative dispersion within each age. Zero-asset shares from age 25 are imposed; the one model outcome is the entrant group, where zero-asset entry puts the share at 44.4% against 30.0% in the SCF. Two dimensions of the distribution are not fitted. Figure 2 shows the first, the age shape of mean wealth. The second is wage dispersion by age. The hourly wage is $w_t h$, so within an age group wage dispersion equals skill dispersion, and only the entrant value is a target. In the CPS the log standard deviation of hourly wages rises from 0.493 at ages 20–24 to 0.68 at ages 50–54. The model imposes the entrant dispersion at every age, so its profile is flat at about 0.53 and understates midlife wage inequality. Cohort averages, which the compensation targets, are little affected; any count of losing households within a cohort should be read as a lower bound.

After 2010, inheritances tie wealth accumulation to the estates of the deceased. Estates are distributed across recipient ages in fixed shares taken from the SCF and equally within each age. The estate-flow target in Table 1 sets the strength of the bequest motive that generates them.

Every reform starts from the same state in 2026: the baseline's distribution of assets and skills by household and its capital, undistributed estates, public debt, and trust-fund reserves, which Table 2 reports next to their 2010 values. Before 2026 households expect the baseline, with scheduled benefits while the fund can pay them, funding-linked cuts after reserve exhaustion, no Treasury guarantee, the statutory payroll rate, and the consumption tax that a 100% debt ceiling requires. A reform is announced unexpectedly in 2026, after which households have perfect foresight about it. Preferences and technology are the same under every policy, so common initial conditions isolate the effects of the benefit rules and their financing.

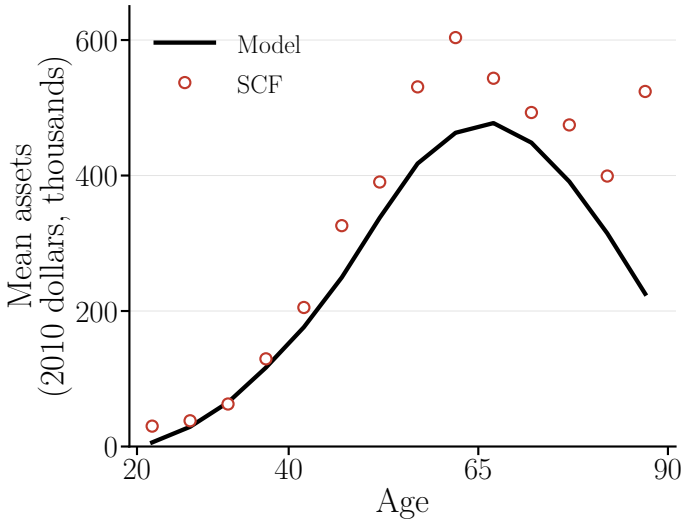


Figure 2: Mean wealth by age at the start of 2010

Notes. Mean nonnegative wealth per adult in the 2010 SCF and mean model assets. The level of model wealth is scaled to capital plus public debt, so only the age shape is a model outcome. Points are five-year age groups at their midpoints; model means use 2010 population weights. SCF family net worth is split equally between spouses, with non-positive values set to zero.

3.4 Fiscal calibration

The 2010 accounts set the levels of the fiscal block and the Trustees' projections set its paths. Appendix A.5 gives every source and construction. Three levels come from the 2010 program accounts. The benefit coefficient, the annualized average benefit of newly entitled 65-year-olds relative to the previous year's taxable earnings per covered worker, is 0.4826, and a fixed factor maps the model's wage bill per working-age adult into taxable earnings per covered worker. Benefits already in payment follow the observed 2010 profile of beneficiaries and benefits by age. Net contributions at the statutory 12.4% rate imply a taxable-payroll share of 0.512, which turns the statutory rate into the effective rate on model earnings. Government purchases and transfers outside Social Security match the 2010 national accounts, with Medicare taken out of cash transfers and supplied in kind so that each expense enters once.

Two rates close the 2010 budget. The earnings-tax rate of 24.6% balances the government budget of the 2010 stationary equilibrium at its debt-to-GDP ratio: spending plus the primary surplus that services the debt net of the borrowing productivity growth allows, less capital-income tax revenue, divided by labor income. Along the transition this rate stays fixed and the consumption tax enforces the debt ceiling. The consumption floor is 9.7% of 2010 GDP per adult, the federal SSI benefit relative to the average retired-worker benefit, and grows with productivity.

The paths are the Trustees' intermediate projections. The statutory payroll rate stays at 12.4%,

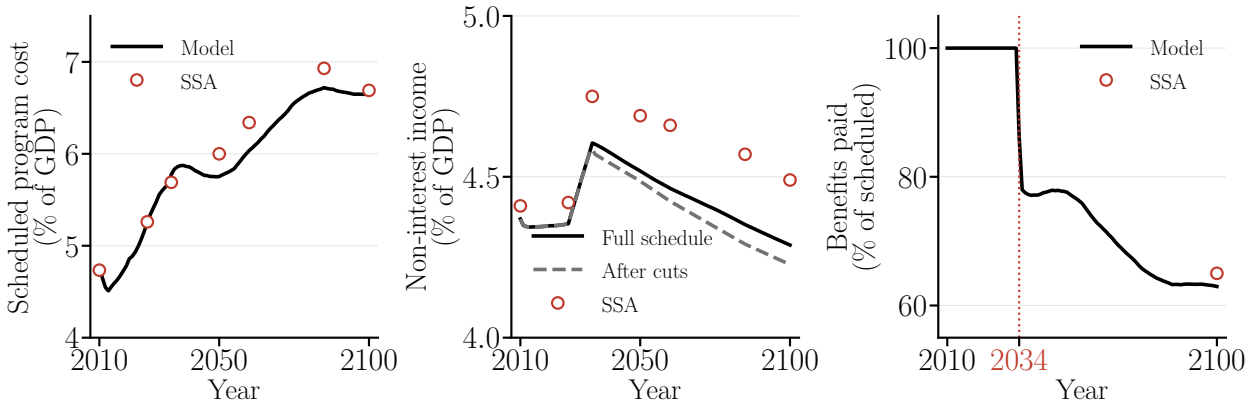


Figure 3: Fiscal calibration: accounts, projections, and model outcomes

Notes. Black lines: baseline. Red circles: 2010 accounts and intermediate projections of the 2026 Trustees Report. Scheduled cost includes gross benefits, administration, and the railroad interchange. Non-interest income includes payroll receipts, taxes on benefits, and specified Treasury reimbursements; the black line assumes full scheduled benefits and the dashed gray line uses benefits paid. The last panel shows the paid share of scheduled benefits, with the Trustees' 2100 projection as the red circle. The vertical line marks reserve exhaustion in 2034. Payroll coverage and benefit composition are projected inputs.

the taxable-payroll share follows projected payroll receipts relative to labor compensation (Social Security Trustees, 2026), and the retired-worker coverage rate and the shares of disability, survivor, and auxiliary benefits follow the decomposition in Appendix A.6. All projected inputs hold their 2100 values thereafter.

Figure 3 compares the baseline with the projections on the three statistics that describe the program's funding pressure: the reserve-exhaustion year, and scheduled cost and non-interest income relative to GDP in 2100. Reserves run out in 2034, the Trustees' year.[10] Scheduled cost reaches 6.65% of GDP in 2100 against 6.69% in the projection. Non-interest income is 4.29% of GDP against 4.49%, and the gap comes mainly from the tax on benefits, which the model applies at a fixed 2.5% while the SSA accounts imply about 5.3% by 2100. The lower income lowers the paid share: the model pays 63.0% of scheduled benefits in 2100 against 65.0% in the projection, and with benefits actually paid, non-interest income is 4.23% of GDP. Because payroll coverage and benefit composition are imposed projections, this comparison measures the fit of the model's demographics, labor supply, and fund accounting conditional on those inputs.

[10]The exhaustion year matches. The within-year reserve balance is not targeted.

3.5 Model fit outside the targets

The calibration targets five household moments and three fiscal statistics. Table 3 reports the untargeted fiscal moments and the zero-asset share of entrants. Appendix Table A.2 adds wages at the six age groups outside the two ratios and wealth at every age.

Table 3: Model fit: untargeted fiscal and entrant moments

Moment	Data	Model
Scheduled cost, 2034 (% GDP)	5.69	5.77
Scheduled cost, 2050 (% GDP)	6.00	5.75
Income, 2034 (% GDP)	4.75	4.61
Income, 2050 (% GDP)	4.69	4.52
Paid share, 2100 (%)	65.0	63.0
Zero assets, ages 20–24 (%)	30.0	44.4

Notes. Data: 2026 Trustees Report intermediate projections and the 2010 SCF. Cost and non-interest income are defined as in Figure 3; income assumes full scheduled benefits. The full comparison is in Appendix Table A.2.

The wage profile is matched at its two anchor points and is slightly less concave than the data in between. The model understates wage growth at ages 25–39, by 8.5% at ages 25–29, and overstates wages at ages 45–59 by up to 4%. With the skill exponent fixed, learning efficiency and depreciation can match the overall rise and the plateau but not the speed of the early rise, so the model has young workers accumulating skill a little too slowly and middle-aged workers a little too fast.

Wealth by age is untargeted apart from its aggregate level. The model reproduces mean wealth at ages 30–44 within about 15% of the SCF. Zero-asset shares from age 25 are imposed from the SCF, so they are not a test of the model. It falls short at the two ends. Entrants hold nothing, so mean wealth at ages 20–24 is $6,000 against $30,000 and too many of them hold zero assets. At ages 45–64 the model is 13–23% below the SCF, and at ages 85–89 it holds less than half of the SCF mean. Model households run assets down in old age because mortality is their only risk and estates are a luxury good, while the SCF's oldest respondents are survivors who are selected on wealth and hold assets against medical and longevity risk the model does not have. The model therefore understates the private wealth of the oldest retirees, which makes Social Security a larger share of their resources than in the data and the late-life benefit supplement of Section 4 correspondingly more valuable to them.

The fiscal block is close to the projections in the years it does not target. Scheduled cost tracks the Trustees' figure at reserve exhaustion and runs slightly below it by 2050. Non-interest income stays below the projection from 2034 on, for the benefit-tax reason given above, and the paid share

in 2100 is correspondingly lower. The baseline therefore carries the same funding pressure as the projection, with a shortfall that is marginally larger because of the lower benefit-tax receipts.

4 Reforming Social Security with Less Losses

4.1 Policy Scenarios

Table 4 lists the three policies, and Figure 4 plots corresponding payroll tax rates and paid shares. All three are announced in 2026 and face the same fiscal rule: the Treasury finances deficits by borrowing until public debt reaches 150% of GDP. The consumption tax then raises the revenue needed to respect the ceiling, while the earnings and capital-income tax rates stay at their 2010 values.

The baseline represents a delayed fiscal adjustment. Tax increases and benefit cuts impose immediate losses on current workers and retirees. Borrowing permits the government to postpone these changes and shift part of the financing burden to later taxpayers. Conflict over who bears the adjustment can delay stabilization, as in Alesina and Drazen (1991). I capture this motive with a rule that preserves taxes and promised benefits while borrowing capacity remains. Hansen and İmrohoroğlu (2016) use a similar assumption: fiscal policy remains unchanged until debt reaches a chosen threshold. Here, 150% of GDP is an imposed limit on borrowing in the policy experiments. The timing rule is taken as given, and households anticipate it after the 2026 announcement.

The baseline keeps the 12.4% payroll tax and the scheduled benefit formula. The Treasury pays full benefits through 2040 to maintain scheduled benefits after reserve exhaustion which is CBO's projections.[11] In this baseline, that support ends when the debt ceiling binds in 2041, and benefits are then limited to what payroll revenue covers. This treats Treasury support as a temporary bridge and restores the link between benefits and program revenue once borrowing room is exhausted.

The rules that apply at the ceiling are assumptions, not model outcomes: Treasury support for Social Security ends, and the consumption tax covers whatever budget gap remains. Other rules are possible, such as raising income taxes or cutting other spending. I choose these model assumptions because they are politically feasible and have been implemented in real policy reforms.

[11] CBO's projection convention assumes adequate funding for scheduled benefits even after trust-fund depletion. It does not give Social Security authority to borrow from the Treasury. General-revenue support after depletion would require legislation; I assume that support as part of the baseline policy. See Congressional Budget Office (2024, Appendix A).

Table 4: Payroll-tax and benefit rules in the three scenarios

Policy	Payroll-tax rule	Benefit payments and financing
Baseline	12.4% in every year.	Pay full scheduled benefits through 2040, using Treasury borrowing after reserves run out in 2034. From 2041, limit benefits to program income: 60.02% of scheduled benefits in 2200.
Main reform	12.4% through 2060; then gradually declines to 3.50% in 2200.	Pay full scheduled benefits through 2040. From 2041, pay partial benefits, with further cuts as the payroll tax falls: 16.06% of scheduled benefits in 2200. Treasury-financed payroll-tax relief and benefit supplements protect transition cohorts (Table B.3).
Phase-out	Gradually declines from 12.4% in 2026 to zero in 2200.	After reserves run out in 2033, pay benefits only from program income; the Treasury does not borrow to cover the program shortfall. Benefits fall to zero in 2200.

Notes. Scheduled benefits are the promised pensions before funding-linked cuts. All policies are announced in 2026. The debt ceiling of 150% of GDP first binds in 2041 under the baseline and main reform and in 2043 under the phase-out, and consumption taxes close the remaining budget gap. Payroll rates exclude cohort-specific relief.

I apply the same rule to all three scenarios, so differences in outcomes come only from payroll taxes, benefits, and how to compensate the cost of transition.

Similar to the baseline, the main reform keeps the baseline rules through 2060 and then lowers the payroll tax in equal annual steps to 3.5% in 2200. Benefits remain tied to payroll revenue, so the paid share of scheduled benefits falls with the tax, to 16% in 2200. On top of this schedule, two cohort-specific instruments, a payroll rate below the economy-wide rate for some cohorts and a benefit above the paid benefit for others, are financed from general revenue.

Lastly, the uncompensated phase-out lowers the payroll tax linearly from 12.4% in 2026 to zero in 2200 and compensates no one. It is the fastest linear removal of the program. Because payroll revenue falls from the first year, reserves run out in 2033, a year earlier than in the baseline, and the Treasury does not borrow to cover the shortfall. Social Security pays what payroll revenue covers from 2034 onward, so the paid share of scheduled benefits is 77% in 2034, 37% in 2100, and zero in 2200. Debt reaches the 150% ceiling in 2043 rather than 2041, since the Treasury no longer finances benefits between 2034 and 2040, and the consumption tax then balances the budget at a lower rate than in the baseline.

4.2 Welfare Measurement

I measure welfare changes in consumption equivalents (CEVs). A CEV is the percentage change in consumption, in every remaining year of life under the baseline, that would make people as

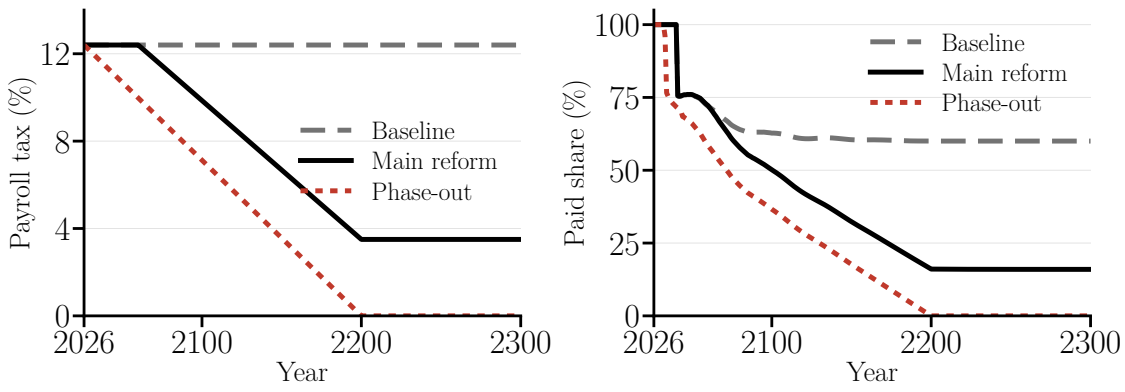


Figure 4: Payroll-tax and benefit paths across the three scenarios

Notes. Payroll rates and benefits paid as a percentage of scheduled benefits, 2026–2300, all under a 150% debt ceiling. The main-reform payroll line is the economy-wide rate before cohort relief; the others are effective rates.

well off as under the reform. A positive CEV is a gain. I report it at two levels:

- *Cohort CEV.* This is the gain or loss of one birth cohort. It averages lifetime utility across the cohort's wealth and skill states, with population weights, and then solves for a common CEV. I evaluate 804 cohorts: the 70 born between 1937 and 2006, who are alive in 2026, and the 734 that enter the economy between 2027 and 2760. A reform leaves *no cohort worse off on average* when the cohort CEV is at or above zero for all 804 cohorts.

- *Aggregate CEV.* This is a single number used to rank policies. It is the uniform consumption change, for all cohorts at once, that matches the policy's change in social welfare, defined below.

For policy rankings, I use a discounted-utilitarian criterion, which sums the lifetime utilities of current and future generations with declining weights on later ones (Calvo and Obstfeld, 1988). The reform is announced in 2026, and utility before the announcement does not change. I therefore count the remaining lifetime utility of each adult alive in 2026 and the lifetime utility of each cohort that enters later, with later entrants discounted by entry date. The household discount factor β governs utility within a life. The social factor β^* governs the relative weights on successive cohorts' lifetime utilities, and it need not equal β (Caplin and Leahy, 2004; Farhi and Werning, 2007). Fleurbaey and Zuber (2015) also distinguish individual discounting within a life from social comparisons between people.

Every adult alive in 2026 has the same per-person coefficient on remaining utility at every β^*, so β^* only moves weight between today's adults and later entrants. This coefficient is a normative choice, and it matters here because the phase-out imposes losses on current retirees while giving

larger gains to later cohorts.[12] However, I do not take a stance on which social discount rate is optimal. Instead, I investigate how different social weights can induce different model implications.

Let $T = 2026$ and $q = \beta^*(1+g)^{1-\sigma}$. Up to policy-independent terms, my main social welfare function for policy comparisons is

$$W(\beta^*) = \sum_{j=20}^{89} \omega_j V_j + \sum_{t>T} q^{t-T} \omega_t V_t, \tag{16}$$

where V_j is the mean remaining-lifetime value of age j at announcement and V_t is the mean lifetime value of an age-20 entrant at date t. The coefficient $\omega_j = n_{j,T} / \sum_{i=20}^{89} n_{i,T}$ is the share of age j in the 2026 model population, where $n_{j,T}$ is the number of people aged j in 2026. The coefficient $\omega_t = n_{20,t} / \sum_{i=20}^{89} n_{i,T}$ is the size of the cohort entering at age 20 in year t relative to that same 2026 model population.

I use $g = 1.13\%$ and $\sigma = 2$. The main reference is $\beta^* = 0.97$, giving a pure annual cohort rate of about 3.1%. At $\beta^* = 1$, there is no pure discounting between entry dates. At $\beta^* = 0$, the function counts current adults only. For a reform, ΔW is equation (16) minus its value under the baseline. The aggregate CEV is the uniform consumption change λ, holding hours and bequests fixed, that solves $\Delta W = \frac{(1+\lambda)^{1-\sigma}-1}{1-\sigma}\mathcal{U}$, where $\mathcal{U}$ is the baseline discounted consumption-leisure term with the same social coefficients. Appendix B.1 derives this equation. This CEV has the sign of ΔW and gives the same ranking against a common baseline. The derivation uses policy differences, so policy-independent utility constants cancel.

4.3 The Baseline Economy

The baseline follows a delayed-adjustment rule for Social Security and its financing (Section 4.1). Figure 5 plots it from 2026 to 2200. The driving force is demography. The number of adults aged 65 and over per 100 adults aged 20 to 64 rises from 32 in 2026 to 45 by 2150, where the population becomes stationary. Benefits already exceed payroll receipts in 2026, 5.2% of GDP against 4.2%, and the gap widens to 1.3% of GDP by 2040. Reserves run out in 2034. The Treasury then borrows to pay full benefits, and public debt rises from 67% of GDP in 2026 to the 150% ceiling in 2041. In that year two things happen at once: the funding-linked benefit cuts begin and the paid share of scheduled benefits falls to 75%, what payroll revenue covers, and the consumption tax, zero until then, jumps to 9.7% to hold debt at the ceiling. As aging continues, the paid share drifts down to

[12] In an overlapping-generations climate model, Kotlikoff et al. (2021) use intergenerational transfers to make current and future generations better off. This provides a related policy-design rationale for transition compensation.

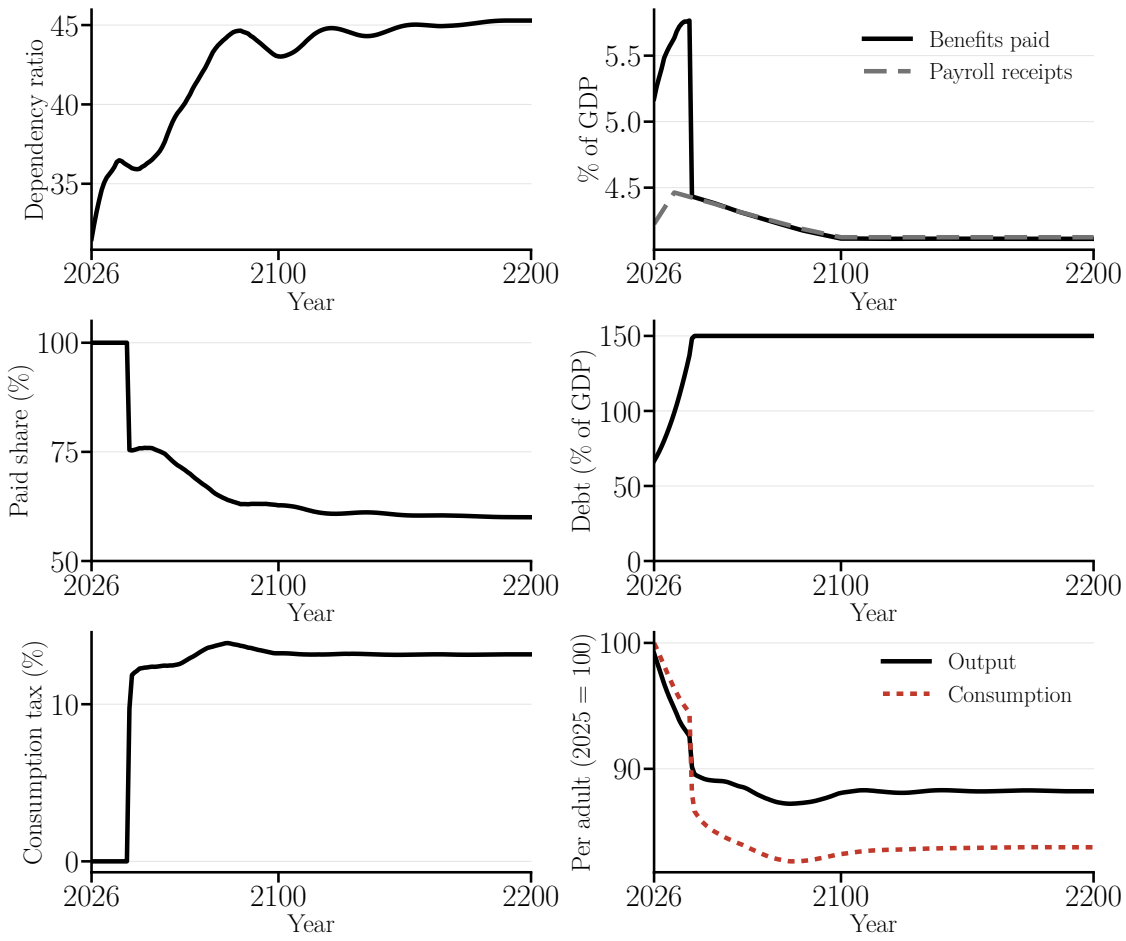


Figure 5: The baseline economy, 2026–2200

Notes. Debt-financed baseline with the 150% ceiling. The old-age dependency ratio is adults aged 65–89 per 100 adults aged 20–64. Benefits paid and payroll receipts are shares of GDP. The paid share is benefits paid over scheduled benefits, on a scale starting at 50. Output and consumption per adult aged 20–89 are indices with 2025 = 100.

72% in 2060 and 60% in 2200, and the consumption tax settles between 12% and 14%.

The economy shrinks per adult without any change in policy. Output per adult falls 10% below its 2025 level by 2041 and 12% by 2080, and hours per adult fall almost as much, for two reasons. The share of adults of working age falls from 76% in 2026 to 69% in 2200, and retirees supply no hours, so hours per adult fall with that share. This accounts for the decline through 2040. In 2041 the consumption tax starts, and hours per worker drop 4% in that one year. A consumption tax pulls hours in two directions: an hour of work buys fewer goods, which makes leisure cheaper and lowers hours, while lifetime resources buy fewer goods, which makes households poorer and raises hours. For a household taxed in every year of its life, the two effects cancel under the Cobb-Douglas composite of consumption and leisure, whatever the curvature σ, and hours are

unchanged.[13]

However, for the cohorts at work in 2041, they do not cancel because much of their lifetime consumption falls before the tax: only the second half of their lives is taxed, and, having seen the tax coming, they bought more goods before 2041 while goods were untaxed. The income effect is therefore weaker than the substitution effect, and these cohorts work less. The size of the drop reflects the low intertemporal elasticity of substitution, one half at $\sigma = 2$: these households cut consumption by less than the tax when it arrives, so their spending in the taxed years rises relative to their wage, whereas a household with an elasticity of one would show no drop at all. As these cohorts retire and are replaced by cohorts that face the tax over their whole working lives, hours per worker return almost to their 2025 level. Output per adult follows hours, with a little extra decline before 2041 because the capital-output ratio decreases from 3.1 to 2.9 while households dissave.

Consumption per adult follows a different path. Until 2040 it holds up better than output, because the Treasury borrows to pay full benefits to a growing number of retirees, who consume out of those benefits while saving falls: investment drops from 20% of output in 2026 to 17% in 2040 and the capital-output ratio from 3.1 to 2.9, so consumption's share of output rises. In 2041 the benefit cut and the consumption tax end this borrowing, saving recovers, and investment returns to 20% of output. At the same time spending on goods outside private consumption, mainly Medicare and out-of-pocket medical care, which rise with age, grows from 17% of output in 2026 to 21% by 2060. Consumption per adult therefore ends 16% below its 2025 level against 12% for output. These declines are the background against which every reform in this paper is measured: the later figures are relative to 2025, so a reform line that falls is not evidence that the reform fails, and a cohort's welfare change is relative to this path, on which retirees after 2041 already receive a quarter less than scheduled. The 150% ceiling is above the 100% limit that households expected before the announcement. It is what lets the Treasury carry scheduled benefits for seven years after reserves deplete. Under the perfect-foresight assumption, it does not affect their comparison.

[13]The cancellation needs only the composite form. With utility over $c(1-\ell)^v$ raised to any power, the first-order condition for hours is $v\,c\,(1+\tau^c)/(1-\ell) = (1-\tau^{ss}-\tau^G)\,w_t h$, so leisure is proportional to spending on consumption divided by the after-tax wage, whatever σ. A consumption tax applied in every year of life scales each year's consumption by $1/(1+\tau^c)$ and leaves spending unchanged, so hours are unchanged; this is the same property that keeps hours constant along a balanced-growth path when wages grow. The curvature matters only across the date the tax starts. For a household at work in 2041, the Euler equation across the jump, holding leisure fixed and with $\beta(1+r) \approx 1$, gives $c_{2041}/c_{2040} = (1+\tau^c_{2041})^{-1/\sigma}$. Only with $\sigma = 1$ does consumption fall one for one with the tax, leaving spending and hours unchanged at the jump; with $\sigma = 2$ consumption falls by less, spending rises relative to the wage, and hours fall.

4.4 The Main Social Security Reform

Here we study a policy reform departing from the baseline in Section 4.3. The primary objective is to find a transition path that will leave no cohorts worse off on average compared to the baseline. In other words, how can we adjust the size of Social Security once the trust fund runs out and the payroll revenue are not enough to cover the scheduled benefit.

Similar to the baseline, the Treasury borrows to pay full scheduled benefits which will hit the debt ceiling in 2041. From 2041, the payroll tax is gradually reduced as well as the benefit until it reaches 16% in 2200. In the reform scenario, none of the 804 cohorts loses on average relative to the baseline, and cohorts born later gain substantially: 3.4% for the cohort born in 2100, and 7.7% in the long run (Table B.1 in the appendix). This subsection explains the result in three steps: who would lose from the cut alone, how the compensation removes those losses, and who pays for it.

4.4.1 Who would lose from the reform alone

The dashed red line in Figure 6 shows cohort welfare when the payroll tax follows the main reform's schedule but no cohort is compensated. How a cohort fares depends on where it is in its life when the cut begins in 2060, and the cohorts fall into three groups.

Older cohorts, born before about 1990, are already retired when the cut begins and are barely affected. They have paid their payroll taxes and collect most of their benefits before the cut reduces payroll revenue. They gain slightly: households that expect lower taxes after 2060 work and save more in the meantime, which enlarges the tax base and lowers the consumption tax.

Transition cohorts, born roughly between 1990 and 2025, are between mid-career and early retirement when the benefit cut begins. They pay the full 12.4% rate for most or all of their working lives, and in retirement they receive benefits that fall with payroll revenue. They pay for the old system and receive the new one. These 35 cohorts lose, but by little: the largest loss, for the cohort born in 2009, is 0.14% of lifetime consumption.

Future cohorts spend most or all of their working lives under the lower tax. They work more under less distortionary taxes, and because work builds skills, they also earn more. Their efficiency gains are large and grow over time. The losses are therefore small and limited to a few decades of cohorts, while the gains are large in the long-run which go to future cohorts. This is the mechanism that allows me to move some amount of welfare from later cohorts to the transition cohorts.

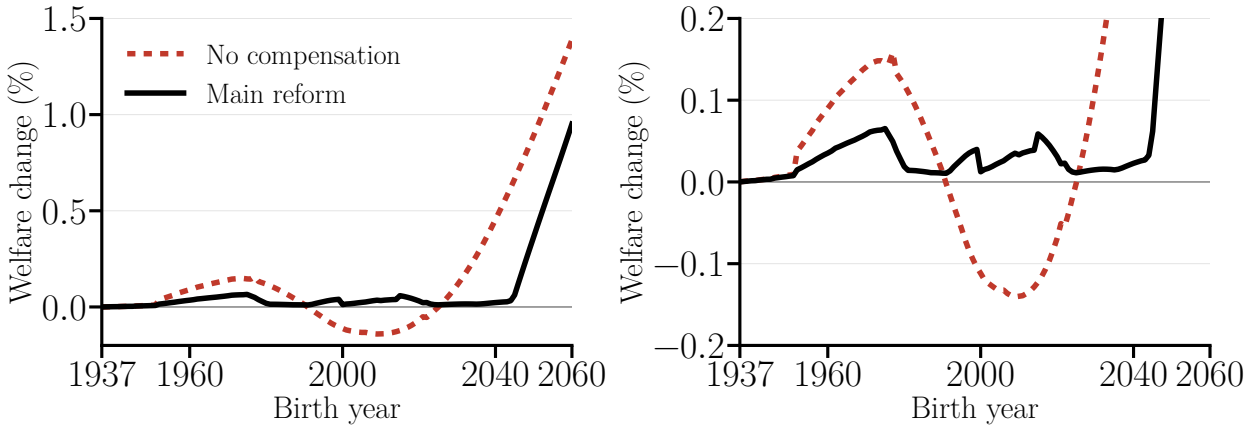


Figure 6: Welfare by birth year under the main reform, with and without the compensation

Notes. Cohort consumption equivalents relative to the debt-financed baseline, births 1937–2060. "No compensation" keeps the main reform's payroll schedule, dates, Treasury guarantee of the interim paid share, and 150% ceiling but removes both instruments; it is solved in full equilibrium at a market tolerance of 10^{-4}. The right panel narrows the vertical scale.

4.4.2 How the compensation works

The reform adds two instruments, both set by birth year and both paid by the Treasury. Each compensates one side of the transition cohorts' loss. The *payroll relief* compensates the high taxes: it lowers the rate a specific-cohort pays on its own earnings below the economy-wide payroll rate, from 2065 until the cohort retires. The *benefit supplement* offsets the lower benefits: it raises the cohort's benefit by a fixed percentage. Neither instrument depends on a household's assets, skill, or earnings history, so each changes the rules a cohort faces rather than handing it a lump sum. Because the Treasury pays for both, the trust fund and the paid share of scheduled benefits are the same as without compensation.

Which instrument a cohort receives depends on how much of its working life remains when the payroll relief starts. Cohorts retired by 2065 can be reached only through the supplement. Cohorts near the end of their careers benefit from the relief for only a few years and rely mainly on the supplement. Younger cohorts work under the relief for longer and need a smaller supplement. Neither instrument works alone. With the relief alone, cohorts born between 1991 and 2014 still lose. With the supplement alone, cohorts born between 2004 and 2033 still lose, because a supplement paid late in retirement does not offset a career of high taxes. Table B.3 in the appendix reports the amounts.

The solid black line in Figure 6 shows the result. Each cohort receives just enough to end

slightly above zero.[14] The compensation reaches more cohorts than the 35 that lose from the benefit cut alone. It also covers their neighbors on both sides, born between 1981 and 1990 and between 2026 and 2043. These cohorts gain from the cut alone, but they also pay for the compensation: financing it keeps the consumption tax higher than it would otherwise be. Their gains are smaller than this cost, so if only the 35 losing cohorts were compensated, these neighbors would lose instead. They therefore receive a small amount as well. The extra cost falls mainly on cohorts born after 2043, whose gains are large enough to absorb it. Every compensated cohort, from 1981 to 2043, ends with a small gain of 0.01% to 0.06%.

4.4.3 Who pays for the compensation

The Treasury pays for both instruments from general revenue. Public debt stays at its 150% ceiling from 2041, as in the baseline, so the consumption tax raises the money. In present value, the transfers cost 2.15% of 2026 GDP. Two features keep this cost from creating new losers.

The first is timing. The consumption tax that pays for the compensation falls on everyone alive in the year it is collected. Cohorts born before about 1980 gain very little from the reform, so a higher tax during their lifetimes would make them worse off. The payments are therefore delayed until the relief starts in 2065, and cohorts born in 2000 or later, who receive the largest supplements, receive them only from 2090 or age 75, whichever comes first. The cost then falls on the transition cohorts, who are compensated anyway, and on later cohorts, whose gains are large enough to absorb it.

The second is growth. The consumption tax exceeds its baseline rate by at most 0.45 percentage points, in 2065, and is below it from 2072 onward, at 8.2% in 2200 against 13.2%. Three forces lower it. Output and the consumption base are larger. Government purchases and cash transfers grow with productivity rather than with output, so they shrink relative to the larger economy; Medicare, a fixed share of output, does not. And the interest rate on the public debt falls as capital accumulates, which at a debt of 150% of GDP saves 0.9% of GDP a year by 2200.

The compensation is thus a transfer from later cohorts to the transition cohorts, paid through the consumption tax. Cohorts that gain from the cut alone give up part of their gain: the cohort born in 2045 gains 0.06% instead of 0.66%. The cost fades for cohorts born later, and the cohort born in 2100 gains 3.42% instead of 3.47%.

[14] The amounts are set by iterating on the equilibrium. Each cohort's relief or supplement is raised in proportion to its remaining loss, the full transition is re-solved, and the step is repeated until no cohort loses.

4.4.4 Macroeconomic Adjustment

Figure 7 plots the adjustment, and Table 5 reports the differences from the baseline alongside those under the phase-out. Until 2060 the two economies are nearly identical. Households learn in 2026 that the payroll tax will fall after 2060 and that their benefits will be smaller, and the two pieces of news push saving in opposite directions: a lower future tax raises expected lifetime income and argues for consuming more now, smaller benefits argue for saving more. The net effect is small, hours per adult 0.2% above the baseline in 2060 and capital per adult 0.05% below, and output per adult differs by 0.2%. Both economies decline relative to 2025 over this period for the demographic reasons.

The reform's effects build as the payroll tax falls, through two channels on the labor side. The lower tax raises the after-tax return to an hour of work, and hours per adult rise 1.6% above the baseline by 2080 and 2.8% by 2100. It also raises the return to the skills that work builds, since each hour now adds to future earnings that are taxed less, and the extra hours themselves produce more learning: skill per hour worked is 1.1% above the baseline in 2100 and 4.2% in 2200. Effective labor, hours times skill, is therefore 4.0% higher in 2100 and 13.4% in 2200, about a third of it from skills.

On the capital side, households that expect smaller benefits save more for their own retirement. Capital per adult is 4.9% higher in 2100 and 23% higher in 2200, and the capital-output ratio rises from 2.94 to 3.13 while it stays flat in the baseline. The two channels differ in timing. Hours respond as soon as a cohort's own payroll rate falls, whereas capital is a stock that accumulates over the working lives of the cohorts that save more, so for the first decades effective labor grows faster than capital. The capital-labor ratio is below the baseline until about 2090, and the wage per unit of skill is slightly below it too, 0.3% in 2070. Only after capital catches up does the wage rise, by 0.3% in 2100 and 2.7% in 2200. The interest rate falls from 4.6% to 4.0% over the same period. Output per adult is 4.3% higher in 2100 and 16% higher in 2200. This is enough to reverse the decline in the baseline. Output per adult returns to its 2025 level around 2170 and is 2.7% above it in 2200, whereas in the baseline it stays 12% below. Aging still lowers the share of adults who work, but under the reform the remaining workers work more hours, with more skill and more capital, and this offsets the demographic decline.

Consumption per adult rises with output, 3.8% above the baseline in 2100 and 16% in 2200, with consumption's share of output unchanged at 59% and the investment share one point higher. Consumption per adult recovers to 97% of its 2025 level by 2200, against 84% in the baseline. The gain is spread across the two groups of adults, though unevenly. Workers consume 4.5% more

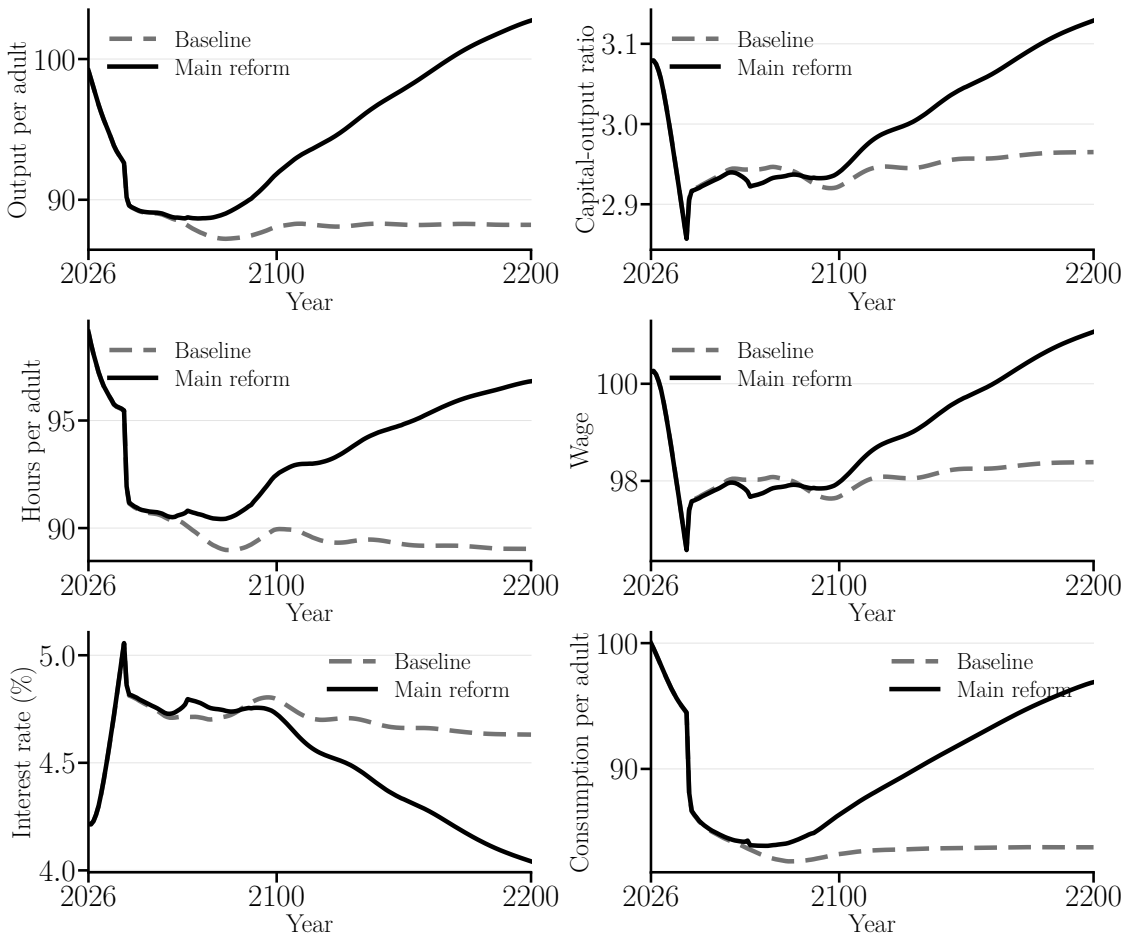


Figure 7: Macroeconomic adjustment under the main reform

Notes. Baseline and main reform, 2026–2200. Output, hours, consumption, and the wage are indices with baseline 2025 = 100, detrended by productivity growth; the first three are per adult aged 20–89, the whole model population. The interest rate is annual, in percent.

than in the baseline in 2100 and 19% more in 2200, as their payroll tax falls and their wage rises. Retirees also consume more, 1.7% more in 2100 and 7% more in 2200, even though they receive much less from Social Security: in 2200, benefits are 1.1% of GDP under the reform against 4.1% in the baseline. Two things make up the difference. These retirees worked under the lower payroll tax and saved more for their own retirement. And they pay a lower consumption tax, 8.2% in 2200 instead of 13.2%. The lower interest rate is the one price that moves against savers, and it is part of why the cohorts that hold assets through the transition gain little on their own and need the compensation. These are the resources that pay for the compensation and lower the consumption tax after 2072.

5 Retaining Social Security versus Phasing It Out

In this section, I study another scenario: the phase-out of Social Security. I compare the main reform with the uncompensated phase-out, both defined in Section 4.1. Table 5 compares the two economies. The phase-out moves at once: the payroll tax falls from 2026, so hours rise, and households expecting smaller benefits save more. By 2040, output per adult is 2.5% above the baseline and capital per adult is 3.6% above. Because the Treasury does not borrow to pay benefits, debt reaches the ceiling in 2043 rather than 2041, so the large rise in the consumption tax comes later, in 2044.

The phase-out's lead in output per adult grows from 2.5 percentage points in 2040 to 7.7 in 2200. In 2200, output per adult is 24% above the baseline under the phase-out and 16% under the main reform, and consumption per adult is 23% and 16% above. More capital per unit of effective labor raises the wage and lowers the interest rate, and the larger tax base lowers the consumption tax. Output, capital, hours, and consumption all end higher under the phase-out because the program ends up smaller.

Table 5: Macroeconomic differences from the baseline

	Main reform				Phase-out			
	2040	2060	2100	2200	2040	2060	2100	2200
A. Per adult, % difference								
Output	0.0	0.2	4.3	16.5	2.5	4.9	10.2	24.2
Hours	0.0	0.2	2.8	8.8	1.5	3.1	5.9	12.3
Capital	0.0	−0.1	4.9	22.9	3.6	5.7	13.1	34.7
Consumption	0.0	0.3	3.8	15.7	0.9	4.4	9.5	22.9
B. Prices and taxes								
Wage (%)	0.0	−0.1	0.3	2.7	0.5	0.4	1.3	4.1
Interest rate (pp)	0.0	0.0	−0.1	−0.6	−0.1	−0.1	−0.3	−0.9
Consumption-tax rate (pp)	0.0	−0.1	−1.2	−4.9	0.0	−1.8	−3.4	−6.8

Notes. Each entry is the policy minus the baseline in the same year. Panel A reports percentage differences per adult aged 20–89. In Panel B, the wage is per unit of effective labor and its difference is in percent; the interest rate and the consumption-tax rate are differences in percentage points (pp).

5.1 The Adults Alive in 2026

However, higher aggregates do not mean that everyone gains. Figure 8 follows the people alive in 2026 over the rest of their lives, relative to the baseline. Under the main reform, consumption

of those already retired barely moves as well as the consumption of working age. Under the phase-out, retirees consume 1.6% less than in the baseline already in 2026, before any benefit cut, because they expect it, and 2.4% less in 2034. Workers consume more from the start, 1.0% more by 2040. Both lines jump in 2041, when the baseline cuts benefits and its consumption tax rises to 9.7%. The phase-out has borrowed less by then, with debt at 129% of GDP, so its consumption tax is still zero. The jump ends in 2044, when the phase-out's consumption tax rises to about 11%: retirees fall back below the baseline, by 1.6% that year and 1.8% in 2050, and workers settle about 2 to 3% above it.

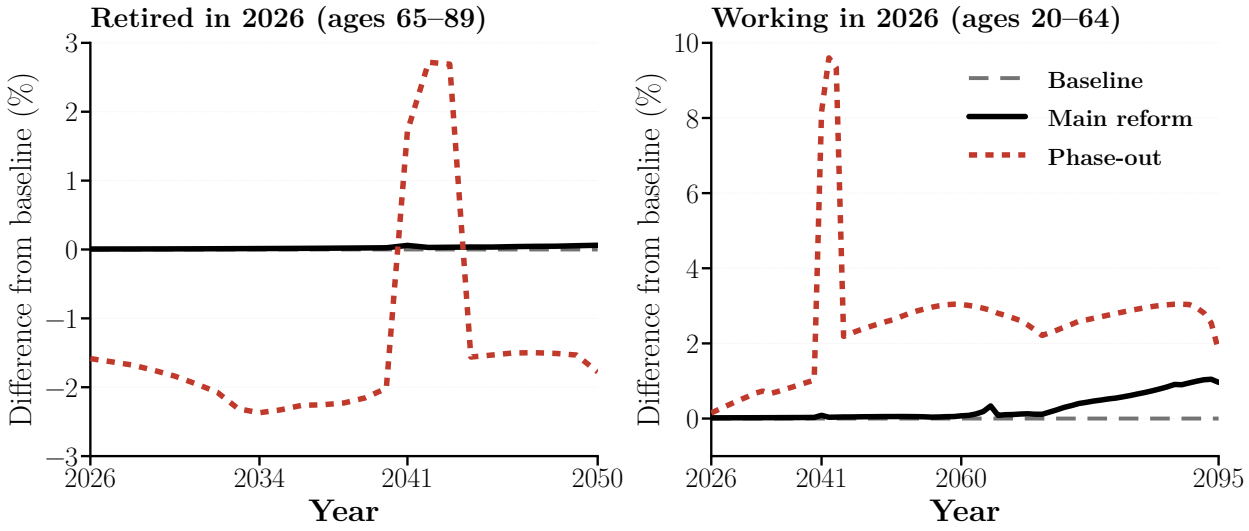


Figure 8: Consumption of the people alive in 2026, relative to the baseline

Notes. The left panel covers those born 1937–1961 and the right panel those born 1962–2006.

These yearly gaps do not settle who is better off, since the jump lasts only three years. Figure 9 therefore takes a lifetime view. For each age in 2026, it reports the cohort CEV: the uniform percentage change in remaining lifetime consumption under the baseline that would make people of that age as well off as under the policy. The black line is the main reform and the red dashed line is the phase-out, and a point above zero means that people of that age gain. The shading shows which policy is better for each age, light red where the phase-out is better and gray where the main reform is better. The size of a shaded area is not a measure of aggregate welfare, because it does not reflect how many people are at each age. The dot at age 0 is not a newborn. It is a single present-value CEV for all future generations together, placed at the left edge of the axis. The two panels differ only in how much weight future generations receive, $\beta^* = 0.8$ on the left and 0.97 on the right. A cohort's own welfare does not depend on this weight, so the lines are the same in both panels, and only the dots and the "Aggregate" labels change. Each label reports the

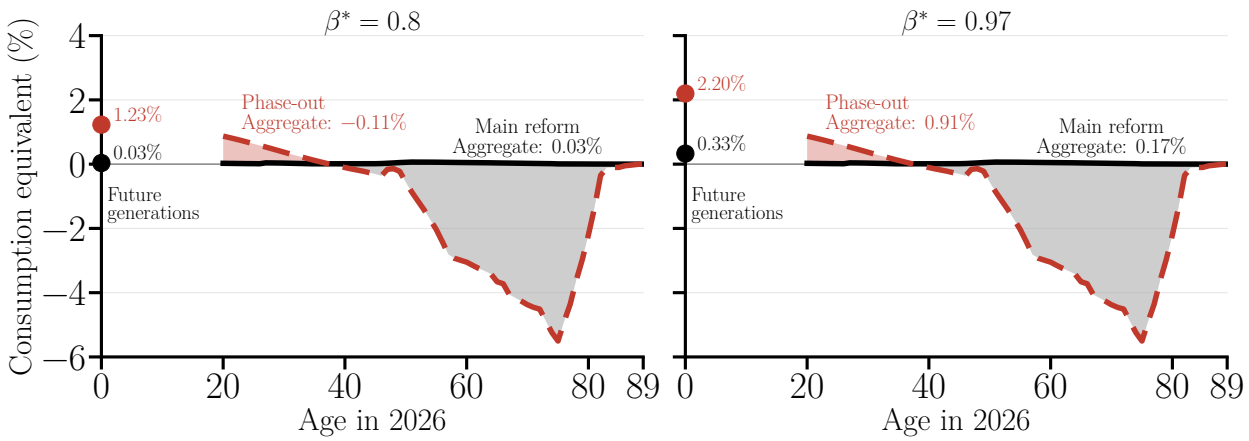


Figure 9: Welfare by age in 2026 and of future generations

Notes. CEVs relative to the debt-financed baseline, in percent. Ages 20–89: cohort-average changes in remaining lifetime consumption, unchanged by the social weights. Shading marks which policy gives the higher cohort CEV; its area is not aggregate welfare. Age 0 is a display position for the uniform CEV of all future age-20 entrants. "Aggregate" is the aggregate CEV, using the weights of equation (16), including the constant tail after 2510. Separate group CEVs do not add. The panels use $\beta^* = 0.8$ and 0.97. The adult-migration welfare term is omitted.

aggregate CEV, which combines current and future cohorts using equation (16), so it is not a sum of the numbers shown. Section 5.3 discusses these aggregate CEVs, and I focus on the lines in this section.

The main reform's curve is flat and lies just above zero at every age, with an average of 0.03%, using the share of adults at each age in 2026, the coefficients ω_j of equation (16). This is because the payroll tax cut does not begin until 2060, and the relief and the supplement hold the transition cohorts at zero, so no current cohort loses on average.

The phase-out's curve crosses zero at age 38. Adults younger than that gain, and the youngest gain the most: the 20-year-olds gain 1% of their remaining lifetime consumption, because the payroll tax they pay falls from their first working year and keeps falling over their careers. Everyone between age 38 and 88 loses. The loss grows with age, to 2.4% for the cohort born in 1970 and 5.5% for the cohort born in 1951. These cohorts have paid the full payroll tax for most or all of their careers, their benefits are cut from 2033, and a loss spread over fewer remaining years is a larger share of each year's consumption. Above age 75 the loss shrinks because fewer years of cut benefits remain. Adults aged 82 or older in 2026 live at most through 2033, the first year of the cut, when the paid share is still 96%, so their losses are close to zero.

In sum, 68% of the adults alive in 2026 are worse off under the phase-out, 41% lose more than 1% of their remaining lifetime consumption, and 27% lose more than 3%. The population-weighted average is a loss of 1.3%. As one uniform change across all these adults, the CEV at $\beta^* = 0$, the

phase-out is worth −0.32% and the main reform 0.03%. Adults aged 62 and over contribute −0.34 points and younger adults 0.02 points to the former. A planner who counts only the people alive today therefore ranks the main reform above the phase-out.

5.2 Benefits, Taxes, and Output

Where do the losses of the adults alive in 2026 and the gains of the cohorts after them come from? Both policies end with a smaller program and a larger economy, so the difference between them lies in the transition. Three things happen at different times under the two policies: benefits are cut, the payroll tax falls, and output rises. Each one affects a different group. The benefit cut reaches the retired, who can do nothing about it. The lower payroll tax reaches the people at work. The extra output reaches the workers who produce it and the retirees they later become. Figure 4 shows the policy rules behind this account. The appendix plots the macroeconomic series in Figure C.2 and the consumption of retirees and workers year by year in Figure C.3.

Benefit cut. The first difference is the benefit cut. For the adults who are retired or close to it, it is the only part of the policy that matters, and their losses in Figure 9 come from it. Under the phase-out benefits fall from 2034; the main reform pays the baseline benefit until 2060. The retirees of 2034 no longer pay the payroll tax, so its fall gives them nothing. They can no longer work more, so the higher return to work gives them nothing either. They respond by drawing down their assets, so their consumption falls by less than their benefits, and the rest of the loss falls on their bequests.

Figure 10 shows why the cut costs these cohorts so much more than the fall in their consumption suggests. Take the cohort born in 1951, aged 75 in 2026, which loses 5.5% of its remaining lifetime consumption. Its consumption, added up over the years it has left, falls by only 2.5%. That average is weighted by dollars, and most of a cohort's consumption belongs to its wealthy members, who barely cut. The welfare measure counts people. Benefits are the same dollar amount for everyone in a cohort, so the cut takes a far larger share of consumption from a retiree with little wealth. Averaged over people, the cohort loses 4.9%. Its bottom wealth tercile loses 6.5%, its top tercile 2.8%, and nearly every household in it loses; the same holds in the cohorts aged 50 to 80. Bequests add the rest. Households value the estate they leave, and the retirees who run down assets to protect their consumption leave smaller estates. For the cohorts aged 65 to 80, the bequest term is between a sixth and a third of the loss, 1.2 of the 5.5 points for the cohort born in 1951. The timing does the rest: these cohorts die before the baseline's own cut in 2041, so they receive none

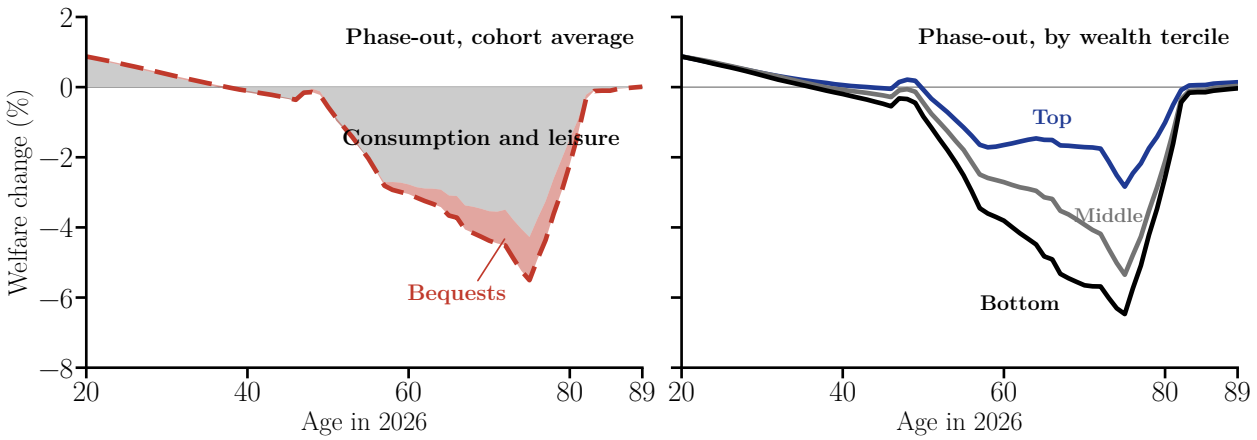


Figure 10: Why the phase-out's losses are large: what the loss is made of and who bears it

Notes. Phase-out against the debt-financed baseline, adults alive in 2026, in percent of remaining lifetime consumption. Left: the cohort-average consumption equivalent, split into a part through consumption and leisure, with the value of bequests at its baseline level, and the remainder through the warm-glow bequest term. Right: household consumption equivalents on the asset and skill grid, averaged within each cohort's wealth terciles. The cohort average values expected utility across households, so in losing cohorts it lies below the person-weighted mean of household values.

of the offset that later retirees get.

Payroll tax. The second difference is the payroll tax. The phase-out lowers it from 2026 while the main reform holds it at 12.4% until 2060. Workers keep the difference, and their consumption rises above the baseline in the first years of the phase-out, while the main reform leaves it unchanged. The consumption tax is lower under the phase-out as well, because the Treasury never borrows to pay benefits and reaches the debt ceiling two years later than in the baseline. In 2200 the consumption tax is 6.4% against 13.2%. The lower consumption tax is the only channel through which the phase-out helps the adults who are retired by 2034, and it arrives after 2040.

Output. The third difference is output. The phase-out moves the economy at once. Hours rise from 2026, because the payroll tax falls from 2026. Saving rises from 2026, because households that expect no benefits save for their own retirement, and the interest rate stays below the baseline from 2048. The main reform is indistinguishable from the baseline until its payroll tax starts to fall in 2060. For the phase-out, its benefit cut comes in 2034 and most of its consumption-tax rise comes in 2044, so hours and consumption fall in two steps. The long run is larger too. In 2200 output per adult is 24% above the baseline under the phase-out and 16% under the main reform, and consumption per adult is 23% and 16% above. Figure C.2 has the full set of series. The extra output goes to the people who produce it. Workers consume more than in the baseline from the first years. The retirees of 2050 and later are different people, cohorts that worked under the lower

tax and saved for their own retirement, and their consumption is above the baseline too. From 2050 on both groups gain. This is why every aggregate series is higher under the phase-out than under the main reform, even though the cohorts that are retired when benefits are cut lose.

This is why the phase-out can dominate the main reform in every macroeconomic series and still fail to be Pareto improving. The two comparisons add up different things. An aggregate series adds up the consumption of the people who are alive in the same year, whoever they are. In every year after the 2040s those people are, on average, better off under the phase-out: the workers pay a lower payroll tax, and the retirees are cohorts that spent most of their careers under that lower tax and saved for their own retirement. Figure 9 instead adds up the consumption of the same person over the years of that person's life. For a cohort that is retired when benefits are cut in 2034, those years are the cut and nothing else. The higher output after 2050, the lower consumption tax after 2040, and the higher wage of the long run all arrive after those retirees have died. Nothing in the later aggregates reaches them.

The two views do not contradict each other. The aggregates say that the phase-out makes the pie larger: output per adult in 2200 is 24% above the baseline, against 16% under the main reform. The cohort view says who gets the slices. The extra output is produced by the workers of the 2030s onward and consumed by them and by the future retirees. The cohorts that are retired when benefits are cut produce none of it and receive none of it. A larger pie is necessary for a transition in which no one loses, but it is not sufficient. It becomes sufficient only if part of the gain is moved to the cohorts that would otherwise lose, which is what the main reform's relief and supplement do and the phase-out does not.[15]

Aggregate output and consumption paths do not by themselves specify a social welfare function. They report outcomes for the people alive in each year. They do not account for utility curvature, leisure, bequests, or the relative weights on cohorts. Under equation (16), the phase-out ranks first when the social factor is close to one: the gains of future entrants receive enough weight to outweigh the losses of today's retirees. The cohort-average Pareto criterion asks whether each cohort, followed over its own life, is at least as well off as under the baseline. The uncompensated phase-out fails this test.

[15]Appendix C.3 shows that the same two instruments make a transition to a payroll tax of zero Pareto improving as well, so the losers of the uncompensated phase-out come from its transition, not from its destination.

5.3 The Role of Social Discount Factor

I now aggregate current and future cohorts using equation (16). The points at age 0 of Figure 9 summarize future entrants, and the labels report the aggregate CEV. The left panel uses $\beta^* = 0.8$, a pure annual cohort rate of 25%; the right uses 0.97, a rate of 3.09%. These are rates on cohort lifetime utility, not consumption discount rates.

At $\beta^* = 0.8$, future entrants gain 1.23% under the phase-out and 0.03% under the main reform. Current cohort CEVs do not move when the weights change. Taken as one group, current adults lose 0.32% under the phase-out and gain 0.03% under the main reform. The aggregate CEV is –0.11% under the phase-out and 0.03% under the main reform. At $\beta^* = 0.97$, future entrants gain 2.20% and 0.33%, respectively. They then receive enough weight for the phase-out to rank higher, with an aggregate CEV of 0.91% against 0.17%. The crossing is $\beta^* = 0.87$, or a pure annual cohort rate of 14.52% as shown in Figure 11. The main reform ranks higher below this factor and the phase-out above it.

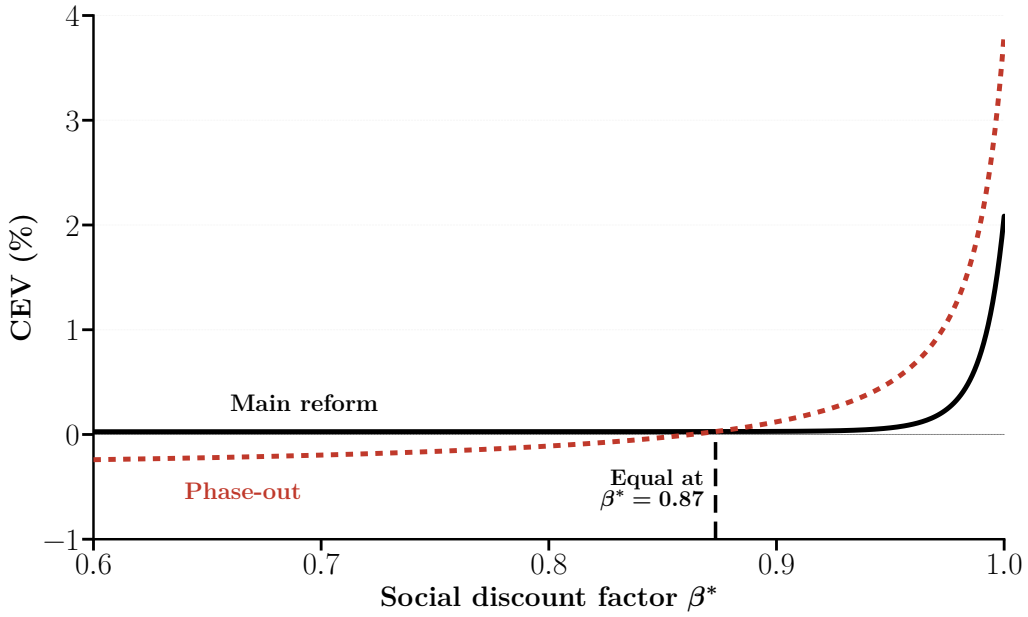


Figure 11: Social welfare by social discount factor

Notes. Aggregate CEVs relative to the baseline, using equation (16), for β^* from 0.6 to 1.

The weights explain the crossing. Adults alive in 2026 keep their population coefficients at every β^*, while a larger β^* gives later entrants more weight (Figure 12). All future entrants together carry 6% of the weight of the 2026 adults at $\beta^* = 0.8$, 13% at 0.9, 38% at 0.97, and 145% at 1. Their share of the baseline consumption term rises from 14% at 0.8 to 49% at 0.97. These are

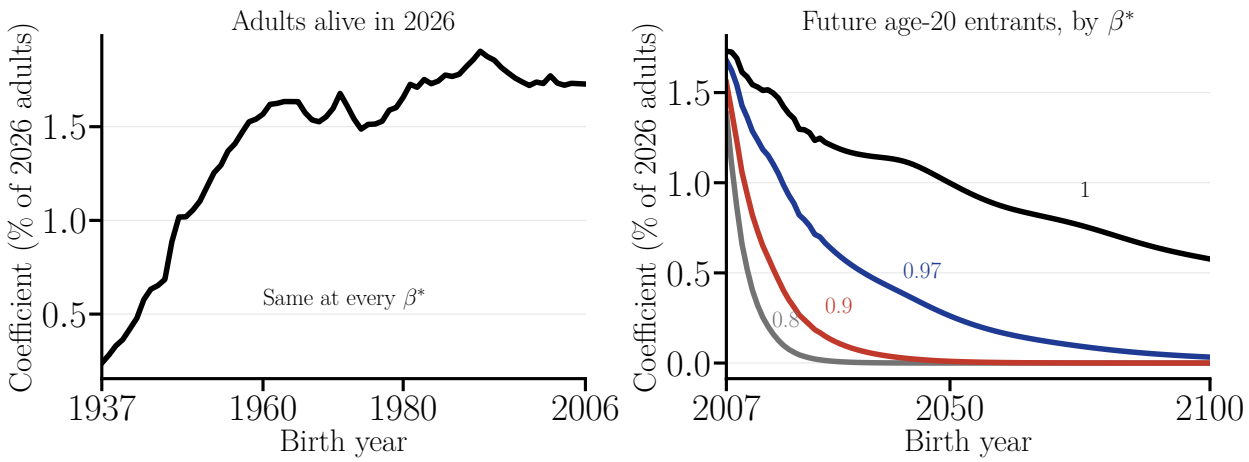


Figure 12: The planner's weight on each generation, by social discount factor

Notes. Coefficients on saved remaining-lifetime utility, as a percentage of the 2026 adult population. Left: the initial population receives its population share ω_j at every β^*. Right: future age-20 entrants receive $q^{t-2026}\omega_t$, with $g = 1.13\%$ and $\sigma = 2$; curves are labeled by β^*. These coefficients do not measure the marginal social value of consumption dollars.

utility coefficients, not marginal dollar values. The gains of the two policies also start at different times. The phase-out reduces payroll taxes from 2026, so its gains start with the first cohort that enters and count at every β^*. The main reform reduces payroll taxes after 2060, so its gains start with the cohorts born after 2045 and count only when β^* is close to one. The main reform gives a small gain to the people alive today. The phase-out gives a larger gain to later cohorts, and today's retirees bear its losses.

The crossing connects the pension comparison to the discounting debate, and I do not identify one ethically correct social discount factor. The appropriate weight depends on whose welfare the government is meant to represent, and the ranking of the two policies changes with that choice. A government that counts only the people alive today sets $\beta^* = 0$. As Section 5.1 shows, the phase-out is then worth −0.32% and the main reform 0.03%.

The dynastic benchmark favors the main reform. The planner treats the next annual entry cohort as the child generation. The same relationship applies to each pair of consecutive cohorts, so generations in the planner's dynasty are $F = 1$ year apart. I use the per-child altruism weight $\alpha_A n^{-\epsilon_A}$ from Córdoba (2015), where n is the number of children per adult and ϵ_A governs how the weight per child falls as n rises. With $n = 1$, $\alpha_A = 0.11$, and the model's detrended household factor $\widetilde{\beta} = \beta(1+g)^{1-\sigma} \simeq 0.9986$, the annual factor on detrended cohort utility in equation (16) is

$$q = \beta^*(1+g)^{1-\sigma} = \widetilde{\beta}\left(\alpha_A n^{-\epsilon_A}\right)^{1/F} = \widetilde{\beta}\alpha_A \simeq 0.1098.$$

The corresponding pure social factor is $\beta^* = \beta\alpha_A \simeq 0.1111$, which is below the threshold. Relative to the baseline, the aggregate CEV is 0.03% for the main reform and −0.31% for the phase-out.

The discounting literature in environmental and climate economics uses much lower pure rates or high discount factor. It follows two approaches. The prescriptive approach treats the pure rate as an ethical choice. Following Ramsey (1928), who called discounting later enjoyments "ethically indefensible," it holds that a generation's welfare should not count for less merely because that generation lives later; Stern (2007) therefore sets the pure rate near zero, at 0.1%, to reflect only the risk of human extinction. The descriptive approach, defended by Nordhaus (2007), instead chooses the pure rate, together with the other preference parameters, to match observed interest rates and saving. For example, pure discounting rates range from 0.1% in Stern (2007) and the 0.5% median in the expert survey of Drupp et al. (2018), through the 1.5% discussed by Nordhaus (2007), to the market-based calibrations of 5.08% in van den Bremer et al. (2026) and 5.8% in van den Bremer and van der Ploeg (2021). All of them lie far below the 14.52% threshold, so the phase-out yields higher aggregate welfare at each one. The recent high rates come from related models, and each is calibrated jointly with risk aversion, intertemporal substitution, and growth, so importing one number does not calibrate the pension model to asset markets.

A planner who gives similar ethical weight to lifetime utility across future entry dates can choose a rate at the low end of this range. At zero pure discounting ($\beta^* = 1$), cohort size and productivity growth determine the coefficients on future entrants' detrended lifetime utility. The aggregate CEVs are then 2.08% for the main reform and 3.82% for the phase-out. Therefore, the policy implication differs by setting. A government that counts only today's adults, or that uses the dynastic weight above, ranks the main reform higher. A government that uses a pure rate from the discounting literature, or no pure rate, ranks the phase-out higher on aggregate welfare, although some cohorts lose under it.

6 Robustness checks

This section asks whether the result depends on learning by doing, the bequest motive, compensation financing and speed, or household protection. Compensation is re-solved in the economic and policy variants; failed protection attempts are reported as such. Each economy uses its own baseline and private discount factor. Supporting results are in Appendix D.

6.1 Learning by Doing

The channel that distinguishes this model from McGrattan and Prescott (2017) is endogenous human capital where work builds skills. To see how it contributes, I switch it off: skill accumulation and depreciation are set to zero, each cohort's skill by age follows the 2010 age profile of the main economy, so that the age-wage profile is unchanged, and the discount factor and leisure weight are recalibrated to the capital-output and hours targets (Table D.1). The leisure weight falls from 3.82 to 2.40. Without a skill return to work, households must value leisure much less to supply the same hours.

Without the skill channel, the reform's gains are smaller and arrive later. Hours respond about as much as with learning, 8.2% against 8.8% above the baseline in 2200, but the extra hours no longer raise skills, so effective labor rises 6.5% instead of 13.4% and output per adult 10.8% instead of 16.5%. The long-run welfare gain falls by more than output, to 2.8% of lifetime consumption from 7.7%, because workers give up almost as much leisure as with learning but earn less from the extra hours. And the gains arrive a generation later: under the same 3.5% schedule with no compensation, cohort gains turn positive only for births after about 2075, and the worst loss is five times larger.

Smaller and later gains are not enough to pay for the compensation. Re-solving the compensation leaves 107 cohorts worse off. Protecting cohorts born in 1977–2045 shifts the cost to those born in 2046–2092, who have no gains to offset it. The largest loss is 3.1% for the 2046 cohort. Extending compensation through the 2090 birth cohort shifts the cost further into the future. This reverses the mechanism in Section 4.4.2. With learning, the reform generates gains large and early enough that the consumption tax used to finance compensation falls mainly on cohorts that remain net winners. Without learning, protecting one group of cohorts shifts the cost to the next.

6.2 Bequests and Retirement Resources

The bequest motive affects the result in two ways. It gives households a reason to save even when pension benefits fall. It also generates inheritances that return the wealth of the deceased to the living. These transfers leave households with little longevity risk to insure.

I remove the bequest motive in two economies. The economies differ in how they transfer the assets of the deceased. In both, I set $\vartheta = 0$ to remove the warm-glow term $\mathcal{W}(x)$ from (5). I recalibrate the discount factor, leisure weight, learning efficiency, and skill depreciation to match the same 2010 targets. Demographics, fiscal rules, learning by doing, and the reform remain

unchanged. The discount factor rises from 1.01 to 1.02 to restore the capital-output ratio. The human-capital parameters adjust to preserve the age-wage profile.

In the *annuitized* economy, the assets of deceased cohort members pass to survivors of the same cohort. No inheritances pass between age groups. Private wealth earns a return that depends on survival, as in an actuarially fair annuity. Social Security therefore provides no additional insurance against longevity risk. In the *inheritance* economy, the assets of the deceased return to the living according to recipient age, as in the main economy. Households bear their own longevity risk. Social Security is their only annuity. Appendix Table D.2 compares the three economies. Table D.3 and Figure D.1 in the appendix report the recalibrated parameters, welfare by cohort group, and cohort profiles.

The Pareto result holds without the bequest motive. The main reform leaves no cohort worse off in either economy. The uncompensated phase-out leaves 50 cohorts worse off in the annuitized economy and 51 in the inheritance economy. This pattern is the same as in the main economy (Appendix Table D.2).

The bequest motive changes how much retirees value the annuity that Social Security provides. This effect appears in the losses of cohorts already retired when the uncompensated cut occurs. Under the phase-out, adults aged 62 and over in 2026 lose 3.3% of their remaining lifetime consumption as a group when their private wealth is annuitized. They lose 3.8% in the main economy. They lose 4.1% when they bear their own longevity risk and Social Security is their only annuity. The ranking of the two policies changes at 0.80 in the annuitized economy, 0.84 in the inheritance economy, and 0.87 in the main economy. The crossing is lowest in the annuitized economy, where retirees lose least, but it also reflects each economy's recalibrated cohort values, so it does not isolate the insurance channel.

6.3 Compensation Financing and Implementation Speed

Appendix Table D.4 varies the source of relief funding and the speed of the payroll tax cut. A payroll surcharge on cohorts born in 2044 or later replaces general revenue as the source of relief funding. These cohorts gain most from the reform. No cohort becomes worse off. At $\beta^* = 0.97$, the CEV falls from 0.17% to 0.16%. The surcharge reduces the gains of the cohorts that receive the most weight at this discount factor. Delaying completion of the payroll tax cut from 2200 to 2300 also leaves no cohort worse off. At $\beta^* = 0.97$, however, the CEV falls from 0.17% to 0.10%. Cohorts born after 2035 pay a higher payroll tax for longer. Neither a slower phase-out nor a smoother consumption tax makes the phase-out Pareto improving. A phase-out completed in

2300 still leaves 42 cohorts worse off. Limiting the rise in the consumption tax to two points a year still leaves 51 cohorts worse off.

6.4 Protecting Every Household

The main reform leaves no cohort worse off on average, but 4.3% of adults alive in 2026, low-skill members of the transition cohorts, lose up to 0.17% of consumption. I re-solve the compensation so that every household with positive population weight has a non-negative CEV, keeping one relief rate and one supplement factor per birth year (Appendix Table D.5). The same two instruments protect every household. The macroeconomic paths and the long-run gain are unchanged, and the CEV at $\beta^* = 0.97$ falls from 0.17% to 0.16%.

Protection costs more because the instruments cannot vary with skill: raising a cohort's supplement to protect its lowest-skill members raises it for everyone in the cohort. The largest supplement rises from 16.6% to 22% of the paid benefit, and the present value of Treasury transfers from 2.2% to 3.3% of 2026 GDP. The consumption tax exceeds the baseline by up to 0.75 instead of 0.45 percentage points and falls below it from 2091 instead of 2072. Instruments tied to earnings would protect households more cheaply, but Social Security's own cohort instruments are enough.

7 Conclusion

Once the Social Security trust fund runs out in the early 2030s, it is not clear what will happen. The government could reform the program to keep it at its current size, or it could let the program pay only what payroll taxes cover, with lower benefits. Neither has happened before. This uncertainty matters for evaluating reform, because a reform's welfare gains are measured against what would happen without it. I study the baseline from the experience of France. Facing repeated opposition to reform, French governments have kept their pension system by borrowing to defer its costs. My baseline assumes the United States would do the same: the Treasury borrows to pay full benefits until public debt reaches a ceiling, and only then are benefits cut to what payroll revenue covers. The system survives, but its costs shift to later generations, and the conflict between generations grows.

This paper shows that Social Security can be reformed without welfare losses for the generations that live through the transition. In an overlapping-generations model with a debt ceiling and learning by doing, gradually lowering the payroll tax from 12.4% to 3.5% leaves every birth cohort at least as well off on average as the baseline. Compensating the transition cohorts costs

about 2% of 2026 GDP in present value, paid from general revenue through the consumption tax.

The implication is that reforming Social Security to a smaller program need not set generations against each other. Pension reforms often stall because their losers are alive, and their winners are not yet born. Relative to a no-reform baseline, compensating a limited set of transition cohorts through lower payroll-taxes and higher retirement benefits avoids cohort-specific losses, which offers a way to reduce intergenerational conflict.

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Appendix

Financing Social Security Reform without Losses

HongSeok Kim

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A Model and calibration details

This appendix gives the demographic inputs for Section 2 and the parameter, household, and fiscal inputs for Section 3.

A.1 Demographic projections

Figure A.1 shows the population projection and its stationary extension. The figure includes ages 20–99; the model uses ages 20–89.

Net migration. The projected population of an age group can differ from the number of survivors from the previous year. I treat this difference as net migration. In the household problem (5), next year's assets are $a' = \kappa_{j,t} x$ with $\kappa_{j,t} = \pi_{j,t} n_{j,t} / n_{j+1,t+1}$. The numerator is the number of survivors from the current age group and the denominator the size of that group next year, including net migrants. Without migration the two are equal, $\kappa_{j,t} = 1$, and a survivor carries all saving into the next year. With net in-migration, $\kappa_{j,t} < 1$: newcomers bring no assets, so the survivors' saving is pooled across the larger group, which lowers assets per person and preserves the group's total saving.

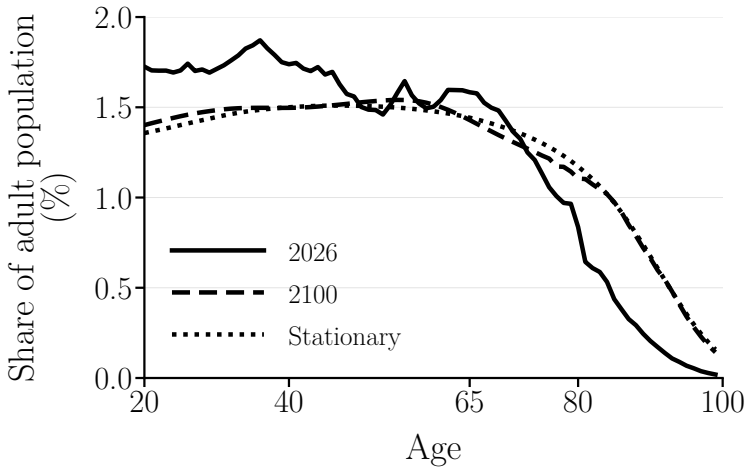


Figure A.1: Projected age distribution of the adult population

Notes. Share of the population aged 20–99 at each age, in percent. The 2026 and 2100 distributions are the intermediate projection of the 2026 Trustees Report. The stationary distribution holds births at their 2100 level and applies the 2099–2100 net survival and migration factors, separately by sex.

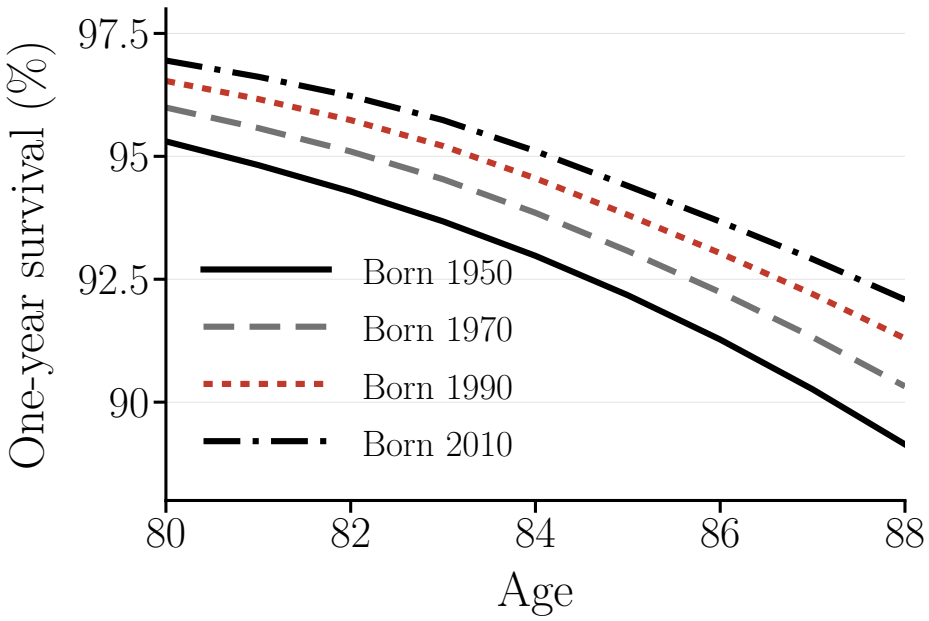


Figure A.2: Annual survival probability by age and birth cohort

Notes. Model survival at ages 80–88 from the intermediate projection of the 2026 Trustees Report, population-weighted across men and women. Cohorts reach the same age in different calendar years. The imposed zero survival at age 89 is omitted.

A.2 Externally assigned parameters

Table A.1 collects the parameters set outside the joint household calibration. Depreciation satisfies $\delta = \alpha/3 - 0.045 = 0.0661$, giving an after-tax return of about 3%.

The bequest scale b_* sets the dollar level at which the bequest motive begins to operate, while the strength ϕ governs how much is left above it. With a single estate-flow target, the two are not separately identified, so I fix b_* and calibrate ϕ. I set $b_* = \$15{,}004$, the average annual retired-worker benefit in 2010, so that households whose resources are close to Social Security income leave little or no estate and bequests come mainly from households well above that level, as in the data. The value annualizes 2009 monthly benefits by single age, weighting each age by its number of beneficiaries (mean \$13,946), and scales them by 1.0759 to match 2010 OASDI outlays (Appendix A.5).

The skill exponent γ governs how the skill gained from an hour of work depends on skill already accumulated: at $\gamma = 0$ the gain is the same at every skill level, and at $\gamma = 1$ it is proportional to skill. The two wage ratios do not identify γ separately from η and δ_h, so I fix it outside the calibration. I set $\gamma = 0.05$, guided by Fan et al. (2024), who estimate the parameter in a Ben-Porath technology in which investment time and existing skill enter jointly. Because that technology differs from (2), the value is not a direct estimate of the learning-by-doing exponent. It implies that the skill gain from work depends only weakly on existing skill.

Table A.1: Parameters set outside the model

Parameter	Value	Description and source or basis
Preferences and technology		
σ	2	Utility curvature
b_*	15,004	Bequest scale (2010 dollars). Average retired-worker benefit in payment in 2010, after the scaling to 2010 OASDI outlays (Social Security Administration, 2010; Social Security Trustees, 2011)
g	0.0113	Productivity growth. Real-wage growth in the Trustees' intermediate projection (Social Security Trustees, 2025)
α	$1/3$	Capital share.
δ	0.0661	Capital depreciation. $K/Y = 3$ (De Nardi et al., 1999; Cagetti and De Nardi, 2006) and a 4.5% pre-tax return
γ	0.05	Skill exponent in learning by doing. Fan et al. (2024)
Social Security		
ρ	0.4826	Retirement benefit at 65 / taxable earnings. 2010 program accounts (Social Security Administration, 2010, 2011)
ζ_{2010}	0.512	Effective taxable-payroll share, using net contributions. Social Security Trustees (2011). Future path from the 2026 Trustees Report
φ_s	0.929	Benefit adjustment for birth cohorts from 1960 onward. Statutory benefit reduction at age 65
τ^b	0.025	Income-tax rate on benefits, credited to the trust fund. Model setting. The 2010 average rate was 3.4%
Taxes, public debt, and the consumption floor		
τ^G	0.246	Earnings-tax rate. Balances the 2010 reference budget
τ^k	0.30	Capital-income tax rate. Kitao (2014)
$\bar{b}$	1.00 / 1.50	Debt ceiling / GDP: calibration and years before 2026 / all policies from 2026 (Section 4.1). CBO projection: debt exceeds 100% of GDP around 2030
$\underline{c}_0$	0.097	Floor / 2010 GDP per adult aged 20–89. 2010 SSI benefit relative to the average retirement benefit (Social Security Administration, 2011)
Health spending		
p_j^M	0.594 / 0.083	Medicare share: ages 65+ / under 65. NHEA tables by age and payer (Centers for Medicare and Medicaid Services, 2012), scaled to 2010 spending
p_j^O	0.14 / 0.22	Out-of-pocket share: ages 65–84 / 85+. NHEA tables by age and payer (Centers for Medicare and Medicaid Services, 2012)
$\mathcal{M}_t$	3.45 / 6.23 / 6.69	Medicare / GDP (%): 2010 / 2050 / 2100. Medicare Trustees (2025)

A.3 Household calibration targets

The five jointly fitted parameters and their targets are in Table 1. This subsection gives the wage and estate calculations behind those targets.

Wages. I use the January–December 2010 CPS outgoing rotation groups. The sample includes workers aged 20–64 with positive hours and earnings. Hourly wages equal weekly earnings divided by usual weekly hours. Mean wages by five-year age group give the two targets: $\$23.20/\$12.18 = 1.905$ for ages 40–44 relative to 20–24, and $\$23.37/\$23.20 = 1.007$ for ages 60–64 relative to 40–44.

Estates. Let n_a be the 2010 population at age a, q_a its SSA death probability, and $\bar{W}_a$ mean SCF wealth per adult. Using the wealth sample in Appendix A.4, the target is

$$\frac{\text{estates}}{\text{GDP}} = \underbrace{\frac{\sum_a n_a q_a \bar{W}_a}{\sum_a n_a \bar{W}_a}}_{\text{share of wealth left by decedents}} \times \underbrace{\frac{K_{2010} + B_{2010}}{Y_{2010}}}_{\text{model wealth / GDP}} = 0.0152 \times 3.53 = 5.378\%.$$

The model counterpart uses the 2010 stationary equilibrium's age profiles with 2010 population weights and the same wealth scaling.

A.4 Initial stocks and household distributions

Units and stocks. With $A_{2010} = 1$, initial effective labor L_0, and $K/Y = 3$, output is $Y_0 = 3^{\alpha/(1-\alpha)} L_0$. The dollar conversion is $D = \$15{,}049 \text{ billion}/Y_0$, using 2010 GDP (Social Security Trustees, 2025). Population is measured relative to adults aged 20–64 in 2010.

Initial capital is $3Y_0$, or $45.147 trillion. Public debt is $0.5298Y_0$, based on $7,811 billion held by the public in December 2009 (Treasury). Trust-fund reserves are $\mathcal{F}_{2010} = 0.1723Y_0$: OASDI assets were $2,592 billion in 2010 dollars (Social Security Trustees, 2011, Table III.A3). Household wealth totals $K_{2010} + B_{2010} = 3.53Y_0$, and the initial undistributed estates is zero.

Wealth and inheritances. The 2010 SCF sample pools respondents and spouses aged 20–89 from 6,482 families. I divide family net worth equally between spouses and set non-positive wealth to zero. I pool the five imputations and divide survey weights by five. Each five-year age group supplies a zero-wealth share and the standard deviation of log positive wealth. These determine a mass at zero and a lognormal positive part. Figure A.3 reports these inputs and the recipient-age shares of inheritances. Each five-year inheritance share is divided equally among its single ages.

Mean initial assets follow the reference age profile, scaled to capital plus public debt:

$$\bar{a}_{j,2010} = \lambda \bar{a}_j^{\text{ref}}, \qquad \lambda = \frac{K_{2010} + B_{2010}}{\sum_{j=20}^{89} n_{j,2010} \bar{a}_j^{\text{ref}}}.$$

This common scaling uses the actual 2010 population and preserves the age profile and relative dispersion. It is separate from the SCF wealth scaling used for the estate target.

Skill dispersion. The standard deviation of log hourly wages among CPS workers aged 20–24 is $s_h = 0.493$, using earnings weights and 13,746 observations. I interpret wage dispersion as skill dispersion and impose a lognormal distribution with this log standard deviation on entrants and all opening cohorts. Mean skill by age follows the 2010 stationary equilibrium.

Asset and skill draws are independent within age because the SCF and CPS samples are not linked. I approximate normal draws with nine Gauss–Legendre nodes per dimension on $[-4, 4]$ and normalize the approximation to preserve means. New age-20 entrants have zero assets.

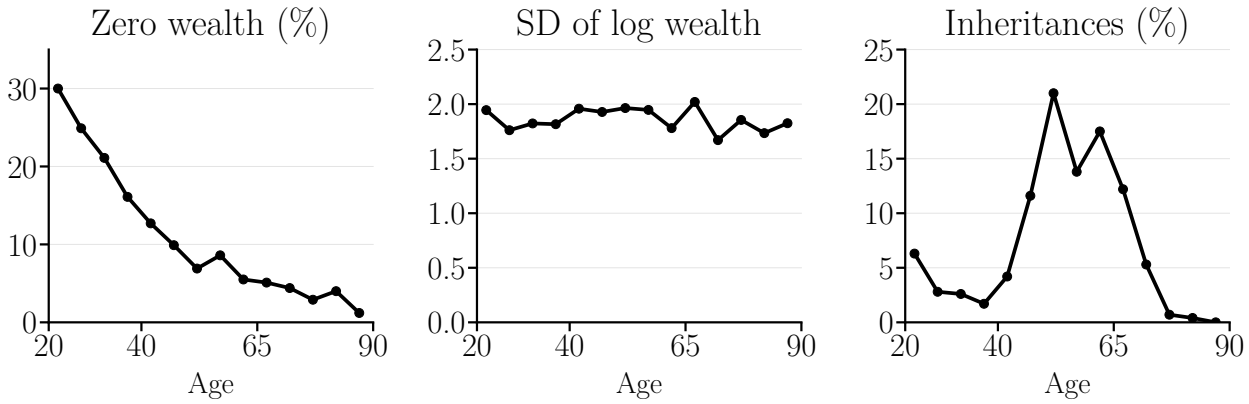


Figure A.3: Distributional inputs from the 2010 SCF

Notes. SCF respondents and spouses aged 20–89, in five-year age groups plotted at their midpoints. Panels show the zero-wealth share, the standard deviation of log positive wealth, and the share of 2010 inheritances by recipient age. Family net worth is split equally between spouses, with non-positive values set to zero.

A.5 Fiscal calibration

Retirement-benefit coefficient. A retiree of age j in year t turned 65 in year $s = t - j + 65$. That cohort's initial age-65 benefit is $\hat{b}_s = \varphi_s \rho \, \bar{e}_{s-1}$, where $\bar{e}_{s-1}$ is taxable earnings per covered worker in the year before the cohort turns 65, ρ is the ratio of the average initial benefit of newly entitled retired workers to these earnings, and φ_s adjusts for the rise in the normal retirement age. The benefit is constant in real terms for life. The coefficient is

$$\rho = \frac{12 \times \$1{,}370}{\$34{,}065} = 0.4826.$$

The numerator annualizes the average benefit of workers newly entitled at age 65 in 2010, at December benefit rates (Social Security Administration, 2011, Table 6.A4). The denominator is 2009 taxable earnings per covered worker in 2010 dollars: \$5,271 billion across 158 million workers gives \$33,375, converted with the CPI ratio $213.967/209.630$ (SSA Annual Statistical Supplement, Table 4.B11).

The model has no taxable maximum or covered-worker count. A fixed factor maps its wage bill per working-age adult into taxable earnings:

$$\bar{e}_t = \psi \frac{w_t L_t}{n_t^W}, \qquad \psi = \frac{\text{2010 taxable payroll per covered worker}}{(1-\alpha) \times \text{2010 GDP per adult aged 20–64}}.$$

Thus, new benefits grow one-for-one with the wage bill per working-age adult. For cohorts born from 1960, $\varphi = 0.867/0.933 = 0.929$ adjusts for the rise in normal retirement age from 66 to 67. Claiming at 65 pays 86.7% rather than 93.3% of the primary insurance amount. I omit the phase-in for births 1955–1959.

Benefits in payment and coverage. Initial benefits use the 2009 retired-worker counts and mean monthly benefits at ages 62–89 (Social Security Administration, 2010, Table 5.A1.1). Grouped counts are split across single ages in proportion to population. A common factor of 1.0759 scales benefit amounts, including disability, survivor, and auxiliary benefits, to 2010 OASDI outlays less lump-sum death payments, \$701 billion (Social Security Trustees, 2011, Table III.A3).

Let ν be the 2010 share of adults aged 65 and over receiving retired-worker benefits. Let ξ_t be beneficiaries per adult aged 65 and over relative to 2010 (Appendix A.6). For $j \geq 65$, the scheduled retired-worker benefit per member is

$$b_{j,t}^R = (1-\tau^b)\,\nu\,\xi_t\,\hat{b}_s, \qquad s = t - j + 65,$$

where $\hat{b}_s$ includes φ_s and τ^b is the benefit-tax rate. Retirees present in 2010 keep their initial benefits. Reform payments are $\chi_{j,t}\phi_t(b_{j,t}^R + b_{j,t}^O)$, as in Section 2.4. The solver carries a multiplier on the retired-worker component of scheduled spending, applied to existing retirees and new beneficiaries alike; it equals one under every policy in this paper and is omitted from the text.

Payroll coverage. The taxable share ζ_t accounts for uncovered work and earnings above the taxable maximum, \$106,800 in 2010 and \$184,500 in 2026. From payroll receipts $\mathcal{P}_t$, I compute

$$\zeta_t = \frac{\tau_t^{ss}}{0.124}, \qquad \tau_t^{ss} = \frac{\mathcal{P}_t}{(1-\alpha)Y_t}.$$

Net contributions of \$637.283 billion imply $\tau_{2010}^{ss} = 0.0635$ and $\zeta_{2010} = 0.512$. The one-time general-fund reimbursement of \$2.4 billion enters separately. The retained SSA comparison

includes the rounded reimbursement in its payroll benchmark, implying $\zeta_{2010} = 0.514$. The model-to-SSA payroll ratio is therefore about 0.9962 throughout the projection.

For 2026–2100, the effective rate is 0.124 times the Trustees' taxable-payroll-to-GDP ratio divided by $1 - \alpha$. I interpolate linearly between 2026, 2034, 2050, 2060, 2085, and 2100 (Social Security Trustees, 2026, Tables IV.B3 and VI.G1), and between the 2010 and 2026 anchors. Coverage remains constant after 2100. The path is imposed relative to projected covered workers, matching projected receipts relative to labor compensation. Under reform, the announced statutory rate replaces 0.124 in $\tau_t^{ss} = \zeta_t \times (\text{statutory rate})_t$. Cohort relief reduces the member's own rate; ζ_t retains its projected path.

Income taxes and spending. The capital-income tax applies to $r(K + B)$. The earnings-tax rate balances the reference budget:

$$\tau^G = \frac{G^{\text{purch}}_{2010} + S^G_{2010} + (r_{2010} - g)B_{2010} - \tau^k r_{2010}(K_{2010} + B_{2010})}{w_{2010}L_{2010}} = 0.246.$$

Purchases are federal consumption and gross investment less fixed-capital consumption and Social Security administration: $1,086 billion, or 7.22% of GDP. Other transfers are federal expenditures less interest, Social Security, and purchases: $1,828 billion, or 12.15% of GDP (Bureau of Economic Analysis, 2011, Tables 3.2 and 3.12). The model replaces observed interest payments with the balanced-growth primary surplus, $(r - g)B$.

Each spending item is a fixed 2010 real amount per person in its group that grows with productivity and with the group's projected size. Purchases are per person of any age, children included. Medicare and other retirement transfers (railroad retirement and pension benefit guaranty) go to adults aged 65 and over. Unemployment insurance, workers' compensation, nutrition assistance, refundable credits, and military medical insurance go to ages 20–64. Remaining transfers are equal per adult. Medicare and military medical insurance are removed from cash transfers and supplied in kind. None of these paths depends on the wage or output of the policy being solved, except Medicare, which is a fixed share $\mathcal{M}_t$ of the output of the policy being solved. The debt ceiling is $\bar{b} = 1$ before 2026 and 1.5 in the policy comparisons. It first binds in 2041 in the baseline and in 2043 under the uncompensated phase-out.

Consumption floor. The floor starts at age 65. It is set from the 2010 federal SSI benefit relative to the average retired-worker benefit, $\$8{,}088/\$13{,}968 = 0.579$ (Social Security Administration, 2011). The implemented floor is 9.7% of 2010 GDP per adult aged 20–89 and grows with productivity. It applies in the calibration and before 2026, then is switched off in the main experiments. Keeping it in the main economy changes no result because it never binds (Section 6.2).

Medical spending. NHEA personal health care spending per person in 2010 is $3,628 at ages 0–18, $4,422 at 19–44, $8,370 at 45–64, $16,872 at 65–84, and $32,411 at 85 and over (Centers for Medicare and Medicaid Services, 2012).

Unscaled Medicare shares are 0.07 below age 65 and 0.50 from age 65. A common factor matches 2010 Medicare outlays of $518 billion, or 3.4% of GDP, giving shares of 0.083 and 0.594. Out-of-pocket shares are 0.11, 0.14, 0.13, 0.14, and 0.22 in the five age groups above.

The Medicare path $\mathcal{M}_t$ sums Part B, Part D, and Hospital Insurance spending relative to GDP. Hospital Insurance spending equals its cost rate times its taxable-payroll-to-GDP ratio. I scale the latter from 0.4355 in 2024 using projected OASDI taxable payroll relative to GDP (Medicare Trustees, 2025, Tables III.B7, III.C10, III.D6).

A.6 Projected fiscal paths

The following inputs complete the payroll-coverage path above. Projected ratios and rates retain their 2100 values thereafter.

- *Retired-worker coverage* ξ_t: beneficiaries per adult aged 65 and over relative to 2010. It is 1 in 2010, 0.997 in 2034, 1.012 in 2050, and 1.044 in 2100 (Social Security Trustees, 2025, Table V.C4).
- *Other benefits* $\mathcal{H}_t^O$: disability, spouse, child, and survivor benefits. They equal 1.5% of GDP in 2010 and retain their projected share of scheduled cost and their 2010 age allocation (Social Security Trustees, 2025, Table IV.B1).
- *Trust-fund return* R_t^F: historical effective yields through 2025 and projected yields from 2026, deflated by the Trustees' CPI (Table VI.G1).
- *Other fund outlays* $\mathcal{O}_t$: administration and vocational rehabilitation, $6.58 billion in 2010, grow with population. The railroad interchange, $4.39 billion, grows with the population aged 65 and over (Social Security Trustees, 2011, Table III.A3).
- *Medicare* $\mathcal{M}_t$: 3.45% of GDP in 2010, 6.23% in 2050, and 6.69% in 2100.

A.7 Calibration fit

Table A.2 reports the moments behind the wage, wealth, and fiscal figures in Section 3. The fiscal comparison retains the income and paid-share gaps discussed there.

Table A.2: Calibration fit: data and model

	Wage ratio (20–24 = 1)		Zero assets (%)		Mean wealth ($1,000)	
Age group	Data	Model	Data	Model	Data	Model
A. Household moments by age group, 2010						
20–24	1.000	1.000	30.0	44.4	30	6
25–29	1.442	1.320	24.9	24.9	38	29
30–34	1.649	1.577	21.1	21.1	63	65
35–39	1.880	1.771	16.1	16.1	129	117
40–44	1.905	1.905	12.7	12.7	205	176
45–49	1.916	1.980	9.9	9.9	326	250
50–54	1.915	1.999	6.9	6.9	390	338
55–59	1.919	1.968	8.6	8.6	531	418
60–64	1.919	1.918	5.5	5.5	604	463
65–69	–	–	5.1	5.1	544	477
70–74	–	–	4.4	4.4	493	449
75–79	–	–	2.9	2.9	475	391
80–84	–	–	4.0	4.0	399	315
85–89	–	–	1.2	1.2	524	225

B. Social Security finances, % of GDP

	Scheduled cost		Non-interest income	
Year	Data	Model	Data	Model
2010	4.73	4.78	4.41	4.37
2034	5.69	5.77	4.75	4.61
2050	6.00	5.75	4.69	4.52
2100	6.69	6.65	4.49	4.29
Reserve exhaustion year	2034	2034		
Paid share in 2100 (%)	65.0	63.0		

Notes. Panel A: wage ratios from the 2010 CPS outgoing rotation groups (earnings weights), relative to ages 20–24, and wealth from the 2010 SCF, with family wealth split equally between spouses. Model wage means use stationary population weights and households with positive hours; model wealth uses the opening 2010 distribution. The fitted targets are the 40–44/20–24 and 60–64/40–44 ratios. Zero-asset shares are imposed from age 25; zero-asset entry changes the 20–24 share. Panel B: baseline against the 2010 accounts and 2026 Trustees projection, with cost and income defined as in Figure 3.

B The main reform

This appendix gives the supporting results for Section 4.

B.1 The aggregate consumption equivalent

This subsection derives the equation that defines the aggregate CEV in Section 4.2. The aggregate CEV scales baseline consumption by $1+\lambda$ for every household counted in equation (16) and at every date, holding hours and bequests at their baseline values. Because period utility is homogeneous in consumption, the scaling factor comes out of every sum.

Start with one period. Write $z = [c(1-\ell)^v]^{1-\sigma}$ for the baseline consumption-leisure term. Then

$$u\big((1+\lambda)c, \ell\big) = \frac{(1+\lambda)^{1-\sigma} z - 1}{1-\sigma} = u(c,\ell) + \phi(\lambda)\, z, \qquad \phi(\lambda) = \frac{(1+\lambda)^{1-\sigma} - 1}{1-\sigma}.$$

The gain in period utility is the baseline term z times a factor $\phi(\lambda)$ that is the same for every household, age, and date.

Next, sum over a life. A household's value is the expected sum of period utilities, discounted by β and survival, plus the bequest term. Bequests are held fixed, so the bequest term does not change. Scaling consumption therefore raises the household's value by $\phi(\lambda)$ times the expected discounted sum of z over the same remaining life. Let Z_j and Z_t be the means of these sums, defined like V_j and V_t in equation (16). For later entrants, Z_t is measured in the same growth-adjusted units as V_t, so the factor $(1+g)^{1-\sigma}$ in q applies to both. Weighting by the coefficients of equation (16) gives

$$W^B(\lambda) - W^B = \phi(\lambda)\,\mathcal{U}, \qquad \mathcal{U} = \sum_{j=20}^{89} \omega_j Z_j + \sum_{t>T} q^{t-T} \omega_t Z_t,$$

where W^B is equation (16) in the baseline and $W^B(\lambda)$ is its value with baseline consumption scaled by $1+\lambda$. The constant $-1/(1-\sigma)$ in u appears in both terms and cancels.

The aggregate CEV sets this gain equal to the reform's welfare change $\Delta W = W^R - W^B$. Solving $\Delta W = \phi(\lambda)\mathcal{U}$ gives

$$\lambda = \Big[1 + (1-\sigma)\frac{\Delta W}{\mathcal{U}}\Big]^{1/(1-\sigma)} - 1,$$

which is $\lambda = \Delta W/(\mathcal{U} - \Delta W)$ at $\sigma = 2$. Because $z > 0, \mathcal{U}$ is positive, and ϕ is strictly increasing with $\phi(0) = 0$. Thus λ has the sign of ΔW, and against a common baseline a larger ΔW gives a larger λ, so the CEV ranks policies as W does. At $\sigma = 2$, $\phi(\lambda) < 1$ for every λ, so a finite CEV requires $\Delta W < \mathcal{U}$.

B.2 Welfare across and within cohorts

The no-loss condition applies to cohort averages. Tables B.1–B.2 report cohort and household welfare.

Table B.1: Cohort welfare under the main reform

Measure	Main reform
Evaluated birth cohorts	804
Losing cohorts	0
Smallest cohort CEV (%)	0.00008
Birth year of the smallest gain	1937
Cohort born 2060: CEV (%)	0.95
Cohort born 2100: CEV (%)	3.42
Long-run cohort CEV (%)	7.73
Current adults only: CEV (%)	0.026
CEV (%), $\beta^* = 0.97$	0.175
CEV (%), $\beta^* = 1$	2.085
Peak public debt (% of GDP)	150

Notes. Main reform against the debt-financed baseline at the 150% ceiling. CEV is the cohort-average consumption equivalent; the minimum keeps five decimals to distinguish it from zero. The CEV is in percent, with a constant tail after 2510.

B.3 Compensation and the long-run rate

Table B.3 reports the two instruments as realized in equilibrium. Figure B.1 compares the main reform with the same transition at other long-run payroll rates: 3.5% gives the highest CEV for social discount factors between 0.83 and 0.91, pure cohort rates of about 10% to 21%.

Table B.2: Within-cohort welfare under the main reform and the household-level path

	Main reform, all households (%)			Losing by skill tercile (%)		Household-level path (%)	
Birth year	Mean	Losing	Worst	Bottom	Top	Mean	Worst
Alive in 2026, by birth year							
Born 1937–1960	0.016	0.0	−0.001	0.0	0.0	0.035	0.001
Born 1961–1979	0.051	0.0	−0.017	0.0	0.0	0.019	0.001
Born 1980–2006	0.019	9.1	−0.170	27.3	0.0	0.032	0.001
All adults alive in 2026	0.028	4.3	−0.170	13.0	0.0	0.029	0.001
Entrant cohorts, by birth year							
Born 2010	0.033	0.0	0.018	0.0	0.0	0.012	0.001
Born 2020	0.031	0.0	0.004	0.0	0.0	0.029	0.001
Born 2030	0.015	2.2	−0.029	6.5	0.0	0.057	0.001
Born 2040	0.023	0.3	−0.018	0.9	0.0	0.069	0.001
Born 2050	0.365	0.0	0.273	0.0	0.0	0.085	0.003
Born 2060	0.951	0.0	0.734	0.0	0.0	0.643	0.514
Born 2080	2.243	0.0	1.711	0.0	0.0	2.046	1.566
Born 2130	5.103	0.0	3.647	0.0	0.0	5.093	3.642
Born 2180	7.461	0.0	5.479	0.0	0.0	7.460	5.478

Notes. Household CEVs at each asset–skill node relative to the debt-financed baseline. Adults alive in 2026 start from their inherited states, with population weights across groups, and entrants from their opening distributions. Mean is the population-weighted ex-ante CEV used in the text, Losing the share of households with a CEV below −0.01%, Worst the lowest household CEV. Entry-skill terciles each hold one third of cohort mass, splitting boundary-node mass pro rata. The welfare decomposition matches the solver value functions to 10^{-12}. The last two columns value the household-level path of Section 6.4; its losing share is zero in every group.

Table B.3: Compensation eligibility, timing, amounts, and financing

	Payroll-tax relief	Benefit supplement
A. Eligibility, timing, and size		
Birth cohorts	2001–2042	1981–2040
Paid	From 2065 until retirement	From retirement (born before 2000); from 2090 or age 75, whichever comes first (born 2000 or later)
Largest adjustment	2.29 percentage points lower (born 2001)	8.63% of higher benefit (born 2020)
Last payment	2106	2129
B. Financing of both instruments		
Source	Treasury general revenue, raised by the consumption tax	
Peak Treasury transfer	0.46% of GDP in 2065	
Present value of transfers	2.15% of 2026 GDP	
Consumption tax relative to baseline	At most 0.45 percentage points higher, in 2065; lower from 2072 onward	
Borrowing allowance	Public debt stays at 150% of GDP	

Notes. Payroll relief lists cohorts with peak relief of at least 0.05 percentage points. Supplements are proportional to the benefit otherwise paid. Financing statistics cover Treasury transfers from 2041; present values use the path's interest rates and output. Tax differences are relative to the baseline.

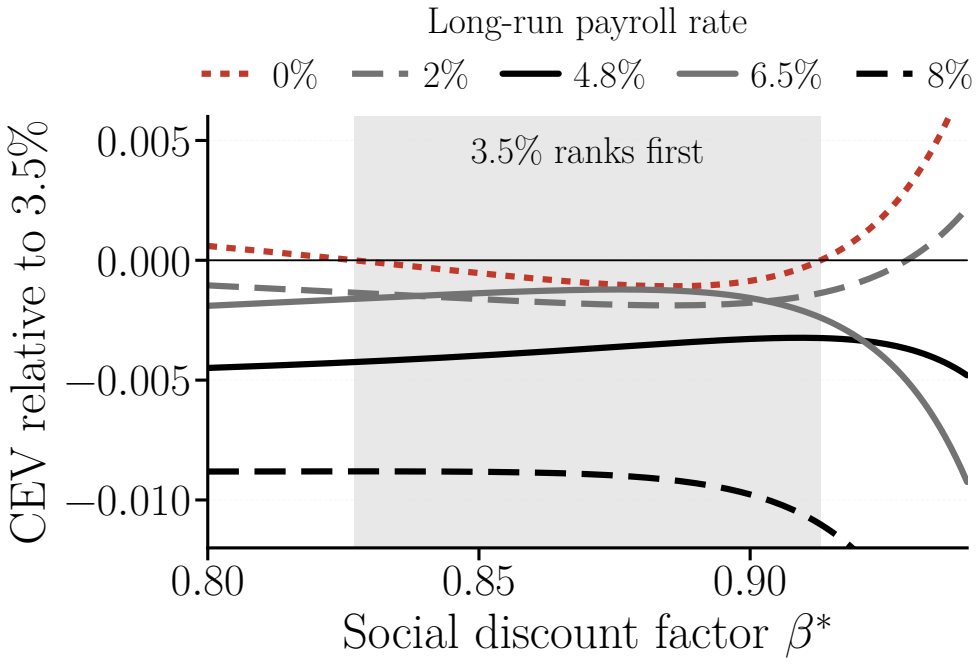


Figure B.1: Social discount factors at which the 3.5% long-run rate ranks first

Notes. Each line is the CEV of a compensated transition with another long-run payroll rate minus the CEV of the main reform, in percentage points, using equation (16). Every transition keeps the main reform's dates and financing, with compensation re-solved so that no cohort loses. The shaded band, $0.827 < \beta^* < 0.913$, is where 3.5% ranks first among the tested rates.

C Phase-out and the long-run payroll rate

This appendix collects the results behind Section 5, in the same order.

C.1 Cohort welfare

Figure C.1 reports welfare by birth year for the cohorts of Section 5.1 and those born later.

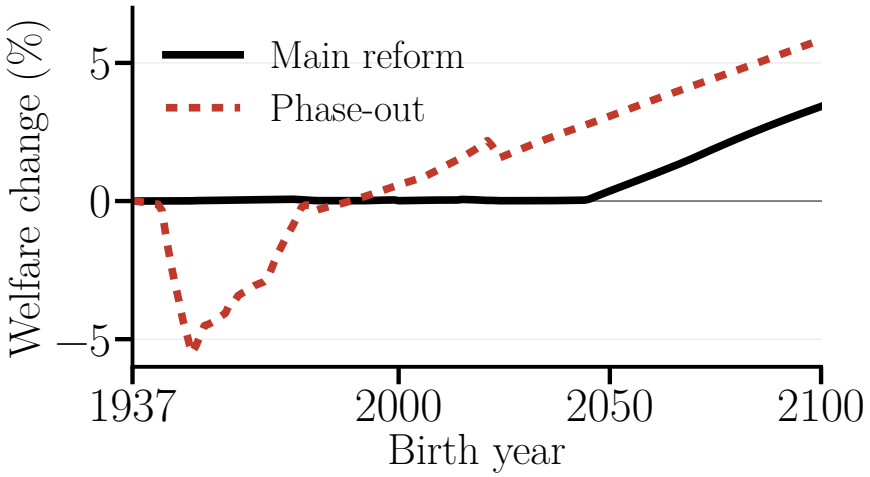


Figure C.1: Cohort welfare under the main reform and phase-out

Notes. Cohort CEVs relative to the common debt-financed baseline, births 1937–2100. Losing-cohort counts use all 804 evaluated cohorts.

C.2 Benefits, taxes, and output

These figures show the paths behind Section 5.2: the macroeconomic series in Figure C.2 and the consumption of retirees and workers in Figure C.3. The policy rules are in Figure 4.

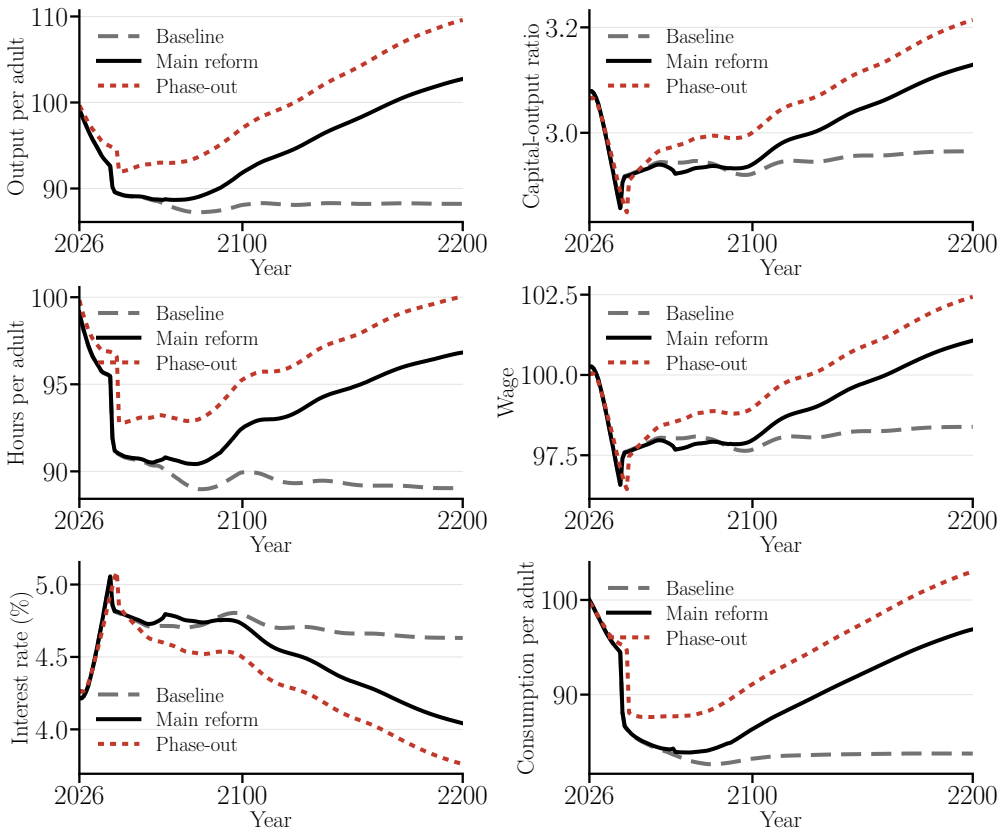


Figure C.2: Macroeconomic adjustment under the main reform and phase-out

Notes. Baseline, main reform, and uncompensated phase-out, 2026–2200. Output, hours, consumption, and wages are indices with baseline 2025 = 100, detrended by productivity growth; the first three are per adult aged 20–89. The interest rate is annual, in percent.

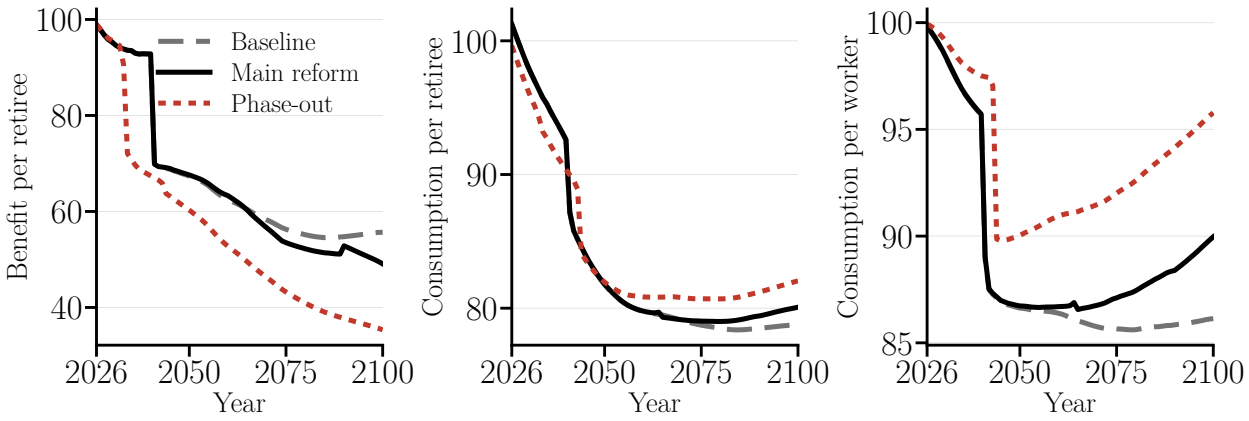


Figure C.3: Benefits and consumption of retirees and workers across the three scenarios

Notes. Indices with baseline 2025 = 100. Retirees are adults aged 65–89 and workers adults aged 20–64. Benefit per retiree is Social Security outlays divided by the number of retirees. Each year's retirees are different cohorts; Figure 9 follows those alive in 2026 over their lives.

C.3 The long-run payroll rate

Table C.1 reports compensated transitions to six long-run payroll rates. No evaluated cohort loses, subject to the stated numerical qualification at 8%. The long-run gain falls from 10.5% of lifetime consumption at a zero payroll rate to 4.0% at 8%. Zero ranks highest among the tested endpoints at $\beta^* = 0.97$ and 1 and for current adults alone, while 3.5% leads by a small margin for factors between 0.83 and 0.91 (Figure B.1). Figure C.4 reports CEV differences from the 3.5% design over $0.85 \leq \beta^* \leq 1$. These are comparisons among solved transitions, not a global optimization over all possible policies.

Table C.1: The same transition at other long-run payroll rates

Long-run rate (%)	Losers	Cohort CEV (%)				CEV (%)		
		Born 2060	Born 2100	Long run	Current adults	$\beta^* = 0.89$	$\beta^* = 0.97$	$\beta^* = 1$
0	0	1.46	4.62	10.52	0.031	0.028	0.239	2.819
2	0	1.17	3.94	8.94	0.027	0.027	0.202	2.405
3.5 (main reform)	0	0.95	3.42	7.73	0.026	0.029	0.175	2.085
4.8	0	0.79	2.96	6.68	0.020	0.025	0.152	1.807
6.5	0	0.46	2.30	5.26	0.021	0.027	0.115	1.408
8	0	0.22	1.71	3.99	0.016	0.019	0.079	1.052

Notes. Each row keeps the main reform's dates, financing, debt ceiling, and instruments and changes only the payroll rate reached in 2200, with relief and supplement re-solved until no cohort loses. At 6.5% and 8% the compensation bands widen to births 1937–2060. At 8% the consumption equivalent of the cohort born in 1937 is -8×10^{-7}, below the numerical resolution, and counts as no loss. All rows use the same debt-financed baseline. CEVs are in percent. The aggregate columns use equation (16), with the adult-migration welfare term omitted; the current-adults column sets $\beta^* = 0$.

Table C.2 answers by splitting the CEV at each rate into the contributions of three groups of cohorts. The first is the adults alive in 2026, who are retired or already at work when the reform is announced. The second is the cohorts born between 2007 and 2035, who enter the labor market between 2027 and 2055 and whose working lives straddle the cut that begins in 2060. These are the cohorts the compensation is built for. The third is the cohorts born after 2035, who spend most or all of their working lives under the smaller program. Each group's contribution is the sum of its members' changes in lifetime utility with the social weights, and I scale these contributions so that they add up to the CEV.

Each group responds differently to the long-run rate. The two rows for the adults alive in 2026 barely move as the rate rises: those who need compensation are held at zero, and those who retire before the cut begins in 2060 gain only through prices. The row for the cohorts born 2007 to 2035 rises with the rate up to 6.5%, because at a higher rate fewer of them fall to the floor and more gain on their own. The cohorts that gain on their own at the higher rate are better off there, while those compensated at both rates are at zero either way. The rows for the cohorts born after 2035 fall as the rate rises, because these cohorts spend their careers under the program, and their

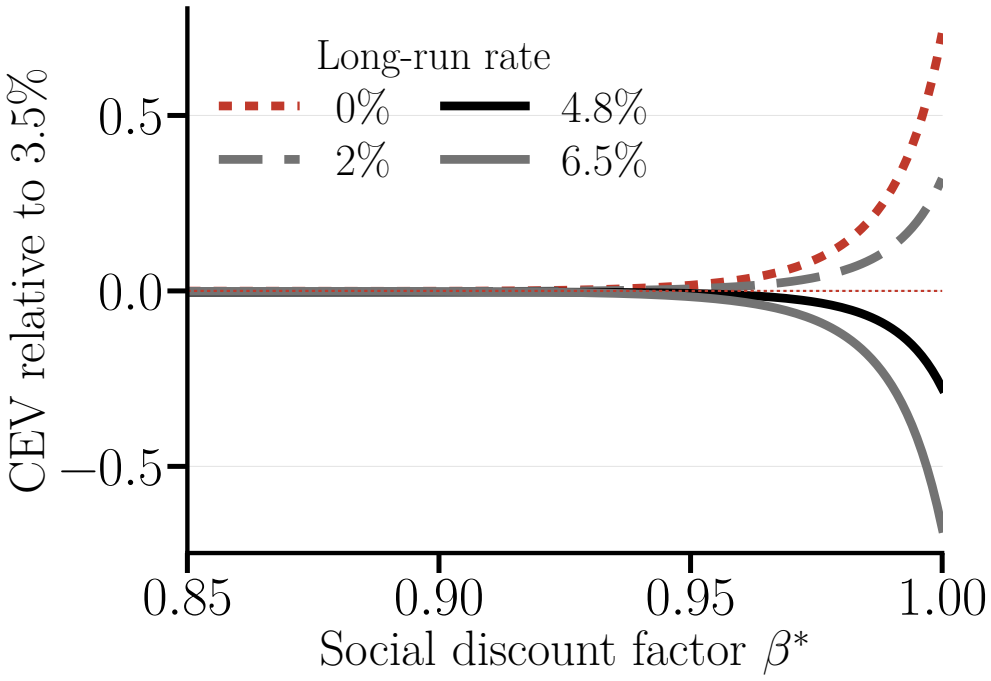


Figure C.4: Social welfare across long-run payroll rates

Notes. Each endpoint uses the main reform's dates and financing, with compensation re-solved. Each line is the CEV relative to the 3.5% rate, which is the dotted line at zero, in percentage points. The horizontal axis shows $0.85 \leq \beta^* \leq 1$. The omitted adult-migration welfare term and the stationary tail follow equation (16).

Table C.2: Who gains from a larger or smaller program

	0	2	3.5	4.8	6.5	8
Social discount factor $\beta^* = 0.87$						
Alive in 2026, age 62 and over	0.003	0.003	0.002	0.001	0.000	0.001
Alive in 2026, under 62	0.021	0.019	0.019	0.015	0.016	0.012
Born 2007–2035	0.002	0.004	0.007	0.008	0.010	0.006
Born 2036–2100	0.001	0.000	0.000	0.000	0.000	0.000
Born after 2100	0.000	0.000	0.000	0.000	0.000	0.000
Total	0.027	0.026	0.028	0.024	0.027	0.019
Social discount factor $\beta^* = 0.97$						
Alive in 2026, age 62 and over	0.002	0.002	0.001	0.001	0.000	0.000
Alive in 2026, under 62	0.013	0.012	0.012	0.009	0.010	0.008
Born 2007–2035	0.004	0.007	0.010	0.013	0.013	0.009
Born 2036–2100	0.158	0.128	0.105	0.089	0.059	0.038
Born after 2100	0.061	0.053	0.046	0.040	0.032	0.024
Total	0.239	0.202	0.175	0.152	0.115	0.079

Notes. Each entry is a group's contribution to the CEV, in percentage points: the CEV times the group's share of ΔW, where that share uses the population-weighted lifetime-utility changes of its cohorts, with initial residents weighted by ω_j and later entrants by $q^{t-2026}\omega_t$. Columns are long-run payroll rates in percent; 3.5 is the main reform. Rows sum to the total.

weight in the sum rises with social discount factor β^*.

The preferred rate depends on how these movements balance. At $\beta^* = 0.87$, moving the terminal rate from zero to 3.5% raises the contribution of cohorts born 2007–2035 by about 0.005 percentage points, and the aggregate CEV rises from 0.027% to 0.028%. At $\beta^* = 0.97$, the same move raises that contribution by about 0.006 points, but the losses in contributions from later cohorts exceed this gain: the aggregate CEV falls from 0.239% to 0.175%. The gains of the transition cohorts therefore decide the ranking only when later cohorts receive little weight.

Where the criterion prefers a positive rate, the reason lies in the transition rather than a larger long-run gain. In this economy earnings and health are deterministic and inheritances return the assets of the deceased to the living, which leaves little longevity risk to insure, so the long-run welfare of future cohorts rises as the program shrinks, as the Aaron condition predicts when the interest rate exceeds the growth rate. The only reason to keep the payroll tax above zero is the cohorts born 2007–2035. They pay the old tax for part of their careers and retire on benefits that are already falling, so a larger cut makes them lose. Compensation brings a losing cohort back to zero and no further, so it adds nothing to social welfare. With a long-run rate of 3.5% instead of zero, the cut is smaller, fewer of these cohorts lose, and some gain on their own, and those gains do add to social welfare.

Positive optimal sizes in the literature come instead from the insurance that Social Security provides against longevity, earnings, and health risk (İmrohoroğlu et al., 1995; Conesa and Krueger, 1999; Huggett and Ventura, 1999; Hosseini and Shourideh, 2019), and from the program's design. Jang et al. (2025) find that the long-run optimum, measured by the steady-state welfare of a newborn, is zero under the current design and positive once the program is made less distortive and more redistributive. The long-run column of Table C.1 is that steady-state measure, and by it the optimum here is zero as well, so these comparisons leave out the insurance value of a larger program. Section 6.2 shows the longevity part of that channel: in an economy without a bequest motive in which households bear their own longevity risk, the uncompensated cut hurts current retirees, the phase-out's worst loss rises to 6.5%, and the crossing rises from 0.80 in the annuitized economy to 0.84 in the inheritance economy.

D Robustness details

This appendix collects the tables and figures behind Section 6, in the same order. Each economy uses its own baseline and recalibrated private factor. Aggregate CEVs use equation (16); current-adults-only entries set $\beta^* = 0$. The adult-migration welfare term is omitted throughout. Compensation attempts with remaining losses are reported as failed attempts.

D.1 Learning by doing

Table D.1 reports recalibration, uncompensated outcomes, and the failed compensation attempts with fixed skills.

Table D.1: The same experiments with and without learning by doing

	Learning by doing	Fixed skills
Recalibrated household parameters		
Discount factor β, before growth adjustment	1.0099	1.0077
Leisure weight v	3.820	2.405
Learning efficiency η / depreciation δ_h	0.451 / 0.0471	0 / 0 (2010 profile fixed)
Uncompensated 3.5% schedule		
Losing cohorts (of 804)	35	100
Smallest cohort CEV (%)	−0.14	−0.74
Cohort born 2100: CEV (%)	3.47	0.40
Long-run cohort CEV (%)	7.73	2.77
CEV (%), $\beta^* = 0.97$	0.185	−0.345
Compensated transition, main instruments re-graded		
Losing cohorts after re-grading (main bands)	0	107
Losing cohorts after re-grading (bands to births 2090)	–	149
Smallest cohort CEV (%)	0.00008	−3.06 (born 2046)
Uncompensated phase-out		
Losing cohorts (of 804)	51	94
Smallest cohort CEV (%)	−5.50	−4.41
Long-run cohort CEV (%)	10.52	3.27
Uncompensated 3.5% schedule in 2200, % relative to the baseline		
Output per adult	16.5	10.8
Hours per adult	8.8	8.2
Effective labor	13.4	6.5
Capital	22.9	20.1
Consumption per adult	15.7	8.3

Notes. The fixed-skill economy sets learning efficiency and skill depreciation to zero, and exogenous age multipliers preserve the main economy's 2010 mean skill profile. The discount factor and leisure weight are recalibrated to capital–output and hours targets, with bequest strength fixed and labor units matching the 2010 moment. Each economy uses its own baseline at the 150% ceiling. Compensation is re-solved for eight rounds with the main bands (supplement births 1977–2043; relief 2000–2045) and ten rounds with both extended to births 2090; fixed-skill rows with remaining losses are failed attempts. The uncompensated schedule keeps the main dates without either instrument, and the long-run block reports its outcomes in 2200. Fixed-skill transitions use a market tolerance of 10^{-4}.

D.2 Bequests and retirement resources

Table D.2 compares the main economy with the two economies without a bequest motive. Table D.3 reports the recalibrated parameters and the contribution of each cohort group to the CEV, and Figure D.1 the cohort profiles in the annuitized economy.

Table D.2: Results with and without a bequest motive

	Bequest motive		No bequest motive			
			Annuities		Inheritances	
	Main	Phase-out	Main	Phase-out	Main	Phase-out
Losing cohorts (of 804)	0	51	0	50	0	51
Smallest cohort CEV (%)	0.00008	−5.50	0.00190	−4.28	0.00208	−6.45
Current adults only: CEV (%)	0.026	−0.321	0.013	−0.170	0.025	−0.256
CEV (%), $\beta^* = 0.97$	0.175	0.906	0.140	0.878	0.140	0.912
Retention preferred for β^* up to	0.8732		0.7994		0.8422	

Notes. Each economy is compared with its own debt-financed baseline under the 3.5% design, with compensation re-solved by cohort. Without a bequest motive $\vartheta = 0$, and the discount factor, leisure weight, learning efficiency, and skill depreciation are recalibrated to the same 2010 targets. Annuities: the assets of the deceased go to their cohort's survivors, and the retirement consumption floor stays. Inheritances: the assets of the deceased go to the living by recipient age, as in the main economy; compensation bands extend to births 2060 and the chain is verified at a market tolerance of 10^{-4}. The last row is the largest β^* at which the retained program beats the phase-out. The CEV is in percent.

Table D.3: Recalibrated parameters and welfare by cohort group

	Bequest motive		No bequest motive			
			Annuities		Inheritances	
	Main	Phase-out	Main	Phase-out	Main	Phase-out
Recalibrated household parameters						
Bequest strength ϑ	150.3		0.0		0.0	
Discount factor β, before growth adjustment	1.0099		1.0177		1.0200	
Leisure weight v	3.820		4.054		3.885	
Learning efficiency η	0.451		0.409		0.402	
Human-capital depreciation δ_h	0.0471		0.0416		0.0407	
Contribution to CEV (%) by cohort group, $\beta^* = 0.97$						
Alive in 2026, age 62 and over	0.001	−0.172	0.001	−0.108	0.000	−0.156
Alive in 2026, under 62	0.012	0.008	0.006	0.026	0.012	0.031
Born 2007–2035	0.010	0.541	0.004	0.489	0.006	0.545
Born 2036–2100	0.105	0.460	0.089	0.408	0.079	0.428
Born after 2100	0.046	0.069	0.042	0.063	0.043	0.065
All cohorts	0.175	0.906	0.140	0.878	0.140	0.912

Notes. Upper block: recalibrated household parameters; other parameters and 2010 targets follow Section 3. Lower block: group contributions to the CEV relative to each economy's baseline, in percentage points at $\beta^* = 0.97$, summing to the total.

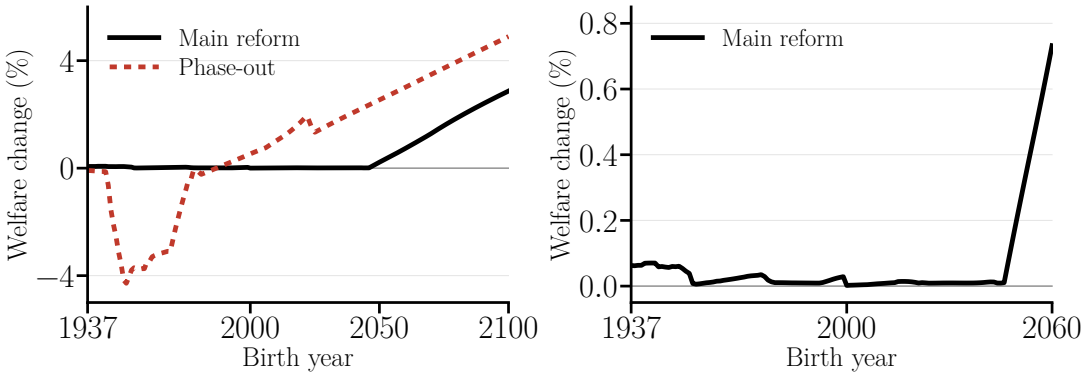


Figure D.1: Cohort welfare in the annuitized economy without a bequest motive

Notes. No-bequest economy with within-cohort asset pooling and the retirement consumption floor. CEVs are relative to that economy's baseline. Left: main reform and uncompensated phase-out, births 1937–2100. Right: main reform, births 1937–2060.

D.3 Compensation financing and adjustment speed

Table D.4 varies compensation financing and adjustment speed. Compensation is re-solved for the main-reform variants. The phase-out variants remain uncompensated.

Table D.4: Alternative compensation financing and adjustment speeds

Policy variant	Losing cohorts	Current adults: CEV (%)	CEV (%), $\beta^* = 0.97$
Main reform	0	0.026	0.175
Relief financed by a surcharge on cohorts born 2044 onward	0	0.023	0.162
Slower main-reform adjustment, completed in 2300	0	0.017	0.101
Phase-out completed in 2200	51	−0.321	0.906
Phase-out completed in 2300	42	−0.194	0.653
Phase-out in 2200 with a smoothed consumption-tax path	51	−0.315	0.921

Notes. All variants use the bequest economy and its debt-financed baseline. Counts cover 804 cohorts; The CEV is in percent. The market tolerance is 10^{-4} for the slower-adjustment case, 5×10^{-5} for the other main-reform rows, and 2×10^{-4} for the phase-out rows.

D.4 Protecting every household

Table D.5 compares the household-level Pareto path of Section 6.4 with the main reform.

The household-level path is valued node by node with the same tool as the main reform. The last two columns of Table B.2 report its mean and worst CEV for each group; no household in any group loses. The search that produced it keeps every numerical limit of the main reform (market residual 5e-5, budget and payroll identities at 10^{-8}) and records an independent replay of the welfare, market, budget, and payroll checks.

Table D.5: The main reform and the household-level Pareto path

	Main reform	Household-level path
Households		
Adults alive in 2026 who lose (%)	4.3	0.0
Worst household CEV, adults alive in 2026 (%)	−0.170	0.001
Worst household CEV, entrants 2030–2200 (%)	−0.029	0.001
Cohort averages		
Losing cohorts (of 804)	0	0
Cohort born 2100: CEV (%)	3.42	3.36
Long-run cohort CEV (%)	7.73	7.73
Current adults only: CEV (%)	0.026	0.029
CEV (%), $\beta^* = 0.97$	0.175	0.156
Instruments		
Payroll relief: birth cohorts	2001–2042	2000–2050
Payroll relief: largest (percentage points)	2.29	2.60
Benefit supplement: birth cohorts	1981–2040	1937–2050
Benefit supplement: largest (% of paid benefit)	8.6	22.0
Benefit supplement: first payment	2090 or age 75 (from retirement if born before 2000)	2026 for cohorts already retired; otherwise as in the main reform
Financing		
Peak Treasury transfer (% of GDP)	0.46 (2065)	0.65 (2090)
Present value of transfers (% of 2026 GDP)	2.15	3.30
Consumption tax above baseline: at most (points)	0.45	0.75
Consumption tax below baseline from	2072	2091

Notes. Both designs use the 3.5% schedule, the 150% ceiling, and the debt-financed baseline. Household rows come from Table B.2: a household loses if its CEV is below −0.01%, and entrants are the cohorts born 2030–2200. The household-level path re-solves relief and supplement on each cohort's worst occupied node until no node in the 804 cohorts is below zero (worst 0.0005%, born 2049), verified by an independent replay at a market tolerance of 5×10^{-5}. Instrument sizes are table values by birth year; relief cohorts have at least 0.05 points. Present values use each path's own interest rates.